

FUGRO

Report of Survey Airborne LiDAR Acquisition Across Tuvalu's Nine Atolls

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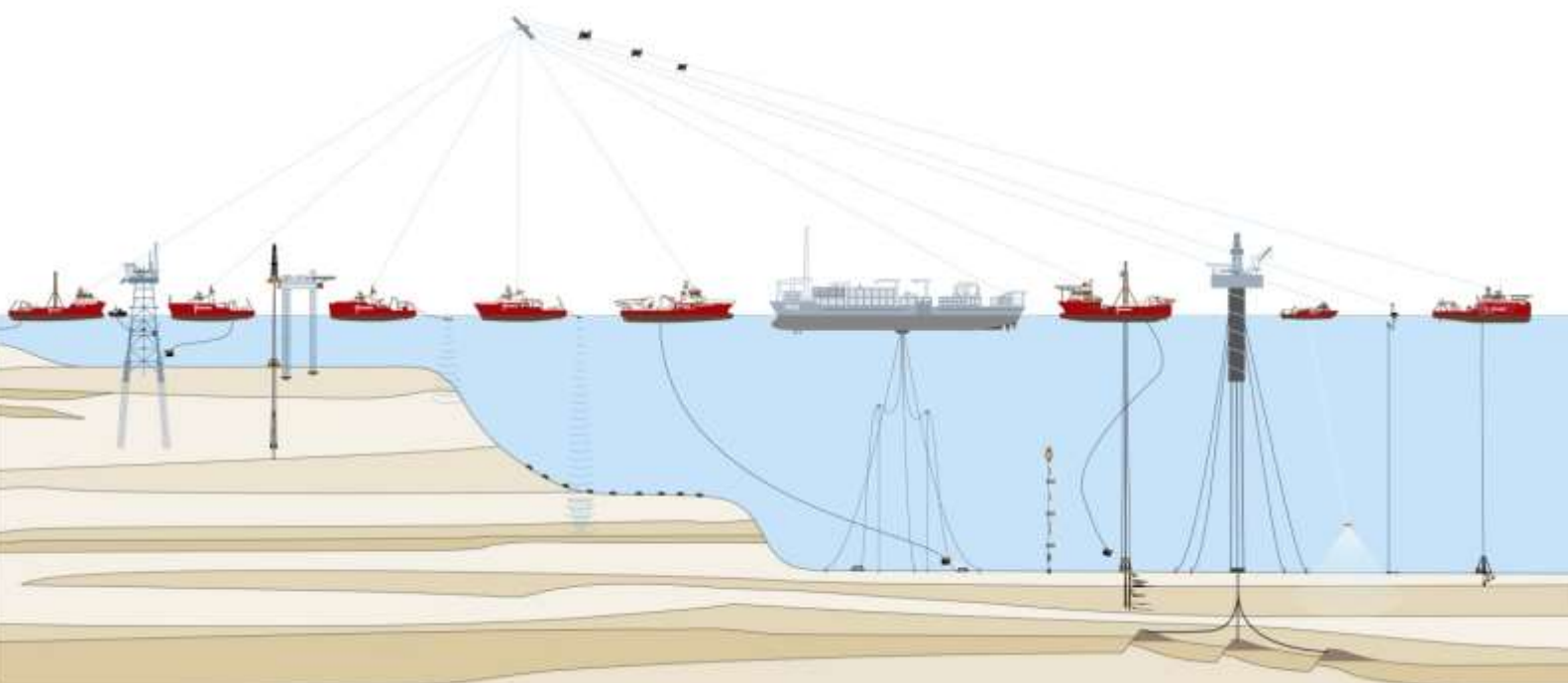
Date of Survey: 22 May - 6 June 2019

United Nations Development Programme



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Version 01 - Draft



FUGRO

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ABBREVIATIONS

AHD	Australian Height Datum
ALB	Airborne LiDAR Bathymetry
AOC	Air Operators Certificate
AS	LADS HD Airborne System
ASC	ASCII grid file format
ASO	Airborne System Operator
ATC	Air Traffic Control
BM	Benchmark
BS	Base Station
CA	Corporate Air
CAD	Computer-Aided Design
CBD	Central Business District
CORS	Continuously Operating Reference Station
DEM	Digital Elevation Model
ECW	Enhanced Compression Wavelet
FAM	Fugro Australia Marine Pty Ltd.
GCP	Ground Control Points
GEOTIFF	GEOspatial Tagged Image File Format
GNSS	Global Navigation Satellite System
GRS80	Geodetic Reference System 1980
GS	LADS HD Ground System
GSD	Ground Sample Distance
IHO	International Hydrographic Organization
IMU	Inertial Measurement Unit
ITRF2014	International Terrestrial Reference Framework 2014
KGNSS	Kinematic Global Navigation Satellite System
LiDAR	Light Detection and Ranging
LADS HD	Laser Airborne Depth Sounder High Density
LAS	LASer file format
LSD	Tuvalu's Lands and Survey Department
MBES	Multi-Beam Echo Sounder
MDD	Mean Depth Difference
MHWM	Mean High Water Mark
NAS	Network Attached Storage
Nd:YAG	Neodymium-Doped Yttrium Aluminium Garnet
NOTAM	Notice to Airmen
PDF	Portable Document Format
POF	Position and Orientation File
POSAV	Position Orientation System Airborne Vehicles
POSPac MMS	Position Orientation System Package Mobile Mapping Suite
PPK	Post Processed Kinematic
PPP	Precise Point Positioning
QA	Quality Assurance



QC	Quality Checking
RGB	Red Green Blue
RTK	Real Time Kinematic
RT-PPP	RealTime - Precise Point Positioning
SA	SubArea
SD	Standard Deviation
SPC	Secretariat of the Pacific Community
STW	Set to Work
TCAP	Tuvalu Coastal Adaption Project
THU	Total Horizontal Uncertainty
TIP	Topographic Integration Points
TVU	Total Vertical Uncertainty
UTM	Universal Transverse Mercator
UNDP	United Nations Development Programme
WADGNSS	Wide Area Differential Global Navigation Satellite System
WSM	Water Surface Model

1. INTRODUCTION

This Report of Survey applies to the project conducted by Fugro Australia Marine (FAM) for the United Nations Development Programme (UNDP). An Airborne LiDAR Bathymetry (ALB) survey and spatial modelling products were required for Tuvalu's nine atolls. The data is intended to support the Tuvalu Coastal Adaption Project (TCAP) Island vulnerability assessment work along with a very broad spectrum of adaptation, development, resource management and environmental monitoring needs in Tuvalu. This data will be integral to assessing the impact of climate change, particularly sea level rise, on the Pacific communities.

The survey aircraft registered VH-VEH (Cessna C441) was mobilised in Goulburn between the 13 to 17 May 2019 with the Fugro LADS HD and RIEGL VQ-820-G ALB sensors. In that period both sensors were successfully ground tested and set to work with two system verification flights. Survey operations commenced with the arrival of the aircraft in Funafuti, Tuvalu and all survey personnel by 23 May 2019. Utilizing Funafuti as the base of operations the aircraft would transit to and from each Atoll, all the way North to Nanumea and Niulakita in the South.

All survey operations were complete after 11 flights, and the aircraft departed Funafuti on 6 June 2019.

Table 1.1 provides an overview of key survey events that occurred during the data collection phase of the project.

Table 1.1: Summary of Survey Events

Date	Event
13 - 17 May	Goulburn - System installation in aircraft VH-VEH and verification flights
20 - 22 May	VEH transit and arrival in Funafuti, Tuvalu
22 - 23 May	FAM Survey Team arrival in Funafuti, Tuvalu
24 May	Field team base set up at Tuvalu's Lands and Survey Department (LSD)
	Survey Operations commence, beginning with Nukulaelae Atoll
30 - 31 May	VEH on standby due to weather
3 June	Survey Operations complete
6 June	VEH and FAM Survey Team depart Funafuti

1.1 Objective

The project objective was to collect and ultimately provide accurate coverage of shallow, near shore and lagoon bathymetry via high resolution bathymetric LiDAR data and spatial modelling products of all of Tuvalu's Nine Atolls. It would also provide LiDAR coverage of all land forms across each atoll, including intertidal reefs and sand flats ephemeral features such as sand banks, etc. The data is intended to support TCAP's island vulnerability assessment work. It will assist in the adaptation, development, resource management and environmental monitoring needs of Tuvalu due to the impact of climate change.

To achieve the best possible data coverage, FAM used the Fugro LADS HD and RIEGL VQ-820-G sensors which operated simultaneously, mobilised on-board Corporate Air C441 aircraft registered VH-VEH. This configuration was successfully used for the 2018 NSW coastal Marine LiDAR survey undertaken by FAM for the NSW Government's Office of Environment and Heritage.

1.2 Survey Area and Strategy

A summary of the initial survey locations awarded is presented in Figure 1.1 and cover the Nine Atolls:

- Nanumea
- Nanumaga
- Niutao
- Nui
- Vaitupu
- Nukufetau
- Funafuti
- Nukulaelae
- Niulakita

Fugro were awarded a contract that originally stipulated data acquisition to occur between the period of 23 March to 13 April, 2019. To lower the risk of poor weather/environmental conditions a Variation Notice was proposed, and a postponement was agreed upon that shifted data acquisition between the period of 23 May to 8 June with a new completion end date of 11 October.

The Tuvalu Lands and Survey Department (LSD) had work underway to occupy strategic benchmarks on all nine atolls (established in the 1970s) and, in partnership with SPC and the UK Hydrographic Office, was to deploy related temporary sea level gauges on each island. This was integral work, conducted in unison to the ALB data collection, to establish the geoid undulation models of each Atoll to the required Mean Sea Level (MSL) datum. These models were supplied to FAM on ?? 2019.

To support the verification of the bathymetric systems an in-country Topographic Integration Points (TIP) was established by FAM and LSD personnel at the start of the data collection period. For more information please refer Annex G.

All sensor calibration, bathymetric LiDAR data collection, turbidity management, data processing, quality control and final product elaboration activities were undertaken by the Fugro team.

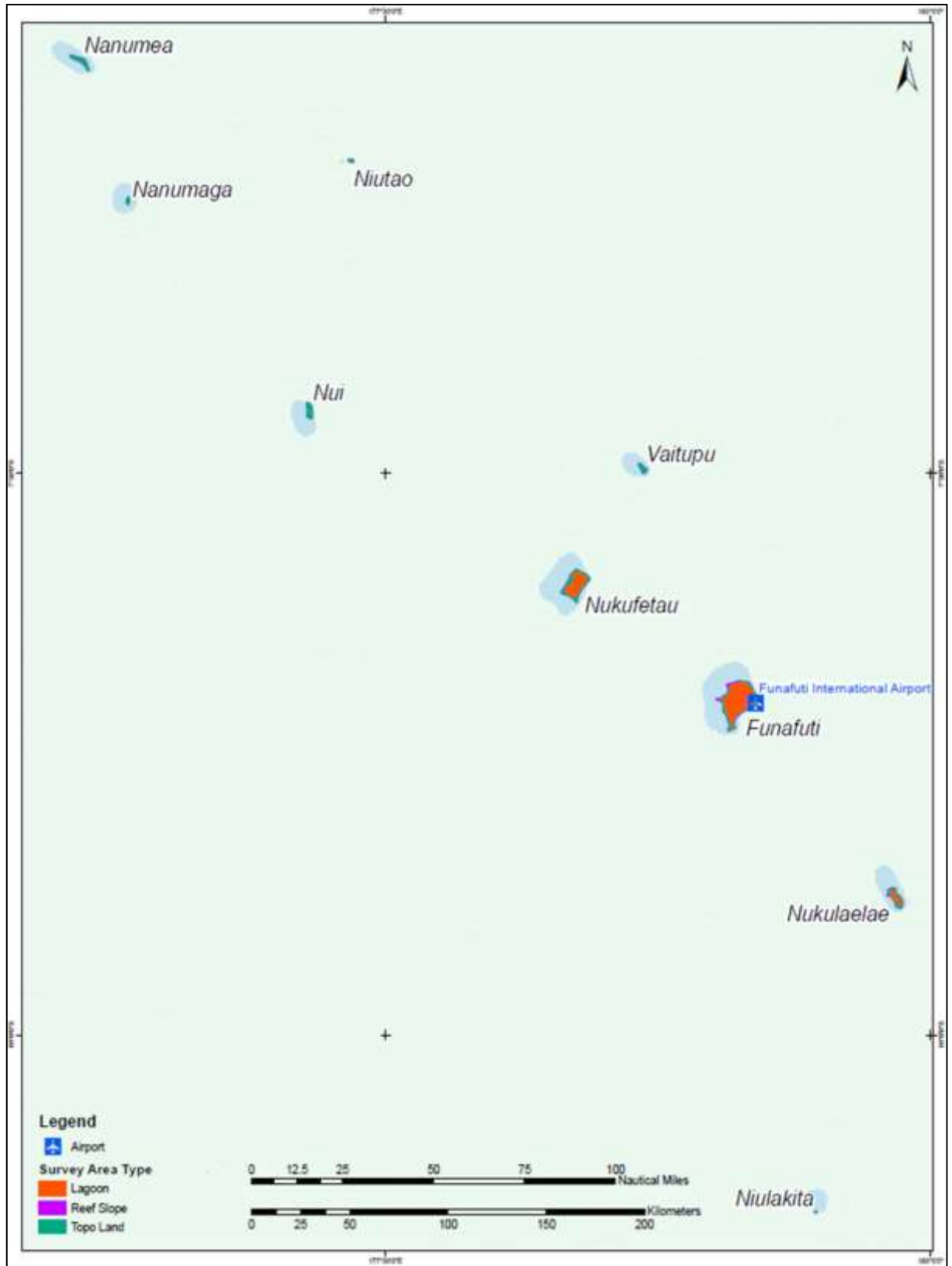


Figure 1.1: Overview of Tuvalu's Nine Atolls

The project team consisted of a partnership established between two main entities:

- i. Fugro Australia Marine (FAM), who:
 - a. Were the contract holder.
 - b. Provided project management, reporting and communication.
 - c. Established the GCP and TIP regions to assess the absolute vertical and horizontal accuracy of the ALB data.
 - d. Performed the ALB system installation, calibration, planning, data collection and processing using the LADS HD and RIEGL VQ-820-G sensors.
 - e. Performed the data processing and QC of LADS HD and RIEGL VQ-820-G datasets in both the Tuvalu field office and Adelaide office.
 - f. Performed processing and QC of the orthophoto mosaics.
 - g. Merged the ALB datasets and created the final digital deliverables which were approved by an IHO Category A hydrographic surveyor.
 - h. Prepared the Report of Survey detailing the results of the project.
- ii. Corporate Air (CA), who:
 - a. Provided the survey aircraft Cessna 441 registration VH-VEH with Certificate of Airworthiness.
 - b. Provided airport/hangar space during ALB systems installation (Goulburn, NSW).
 - c. Provided pilots.
 - d. Provided aircraft maintenance.
 - e. Arranged flight permit for ALB operations.

The ALB systems were operated simultaneously during the data collection phase. The RIEGL VQ-820-G system achieved very high-density coverage at nominally 0.5m spot spacing across topographic areas and shallow water (<7m). The Fugro LADS HD system used a 3.4m spot spacing, collecting data generally to depths between 3m and 40m within the lagoons and 3m to 55m offshore.

During the data collection phase, a field processing office was set up at Tuvalu Lands and Survey Department where data was processed, assessed and then backed up daily in field. This allowed for identification of system errors and areas of poor coverage in order to plan reflights for subsequent flights.

The depth penetration of ALB systems is dependent on several factors including water clarity, sunlight noise, airborne particulate matter and the nature of the seabed. The water clarity across all Tuvalu's Nine Atolls was overall very good with a large amount of coverage achieved within the lagoons. Sparse patches of turbidity were experienced in some lagoons, particularly Funafuti, and reflight with some improvements shown over the course of the survey period.

1.3 Project Specifications / Standards

The project specifications for the data acquisition are outlined in Table 1.2:

Table 1.2: Project Specifications

Requirements:	Accuracy standard:	■ IHO Order 1b for the entire project
	Sounding density:	3 – 5m sounding density. Fugro used the following sounding pattern / density: ■ LADS HD: □ 3.4m x 3.4m spot spacing (scan pattern: H34D18F356) □ 3.2m x 3.2m spot spacing (scan pattern: H30D14F277) ■ RIEGL VQ-820-G: 284 kHz / Nominally ~3.4pts/m²
	Imagery	5cm GSD ortho-rectified mosaic
Geodesy / Datum:	Vertical:	■ Data Collection referenced to the Ellipsoid GRS80 and ■ Final data will be reduced to MSL using a GEOID undulation model for each atoll to be provided by SPC / Tuvalu Lands and Survey Department.
	Horizontal:	■ WGS84 (ITRF2014) ■ UTM60S

All positioning data was post-processed using Precise Point Positioning (PPP) to provide a Post Processed Kinematic (PPK) Global Navigation Satellite System (GNSS) solution which was then applied to all the data.

All ALB data during data acquisition was referenced to the ITRF2014, GRS80 ellipsoid, in UTM coordinates Zone 60 south and heights relative to the GRS80 ellipsoid.

Final ALB data was delivered with the vertical datum reduced to MSL using a GEOID undulation model for each atoll. These were provided by SPC / Tuvalu Lands and Survey Department.

2. SURVEY OPERATIONS

2.1 System Installation and Set to Work

From the period of 13 to 17 May 2019 both the LADS HD and RIEGL-VQ-820-G ALB systems were installed into Corporate Air's VH-VEH aircraft and set to work (STW) checks were performed prior to departure to Tuvalu.

All set to work was conducted over the Wollongong, NSW area where accurate RTK surveyed TIP and GCP data had been collected prior.

A full report of the set to work activities and verification results can be found in the separate report provided, titled "TLC00.066.005-TUV 1 Tuvalu_Set To Work & Verification Report_Rev_3".

2.2 Field Deployment and Planning

Prior to the data collection phase, a number of planning activities occurred in preparation to commence survey operations and continued throughout the data collection phase, as outlined in the following sections.

2.2.1 Flight Permissions

The process of obtaining flight permissions in Tuvalu was very straightforward as the aircraft is an Australian registered aircraft and had a current Air Operators Certificate (AOC) to conduct aerial survey work. This was managed by Corporate Air.

2.2.2 Restricted Airspace Management

During the project planning phase all applicable aeronautical charts were inspected to determine if survey areas crossed or were in close proximity to Restricted, Prohibited or Dangerous airspace zones.

There was no Restricted Airspace within the survey areas.

2.2.3 Survey Areas

The areas of interest for this project, coverage estimated targets and actual coverage achieved statistics are provided in Table 2.1 below.

Table 2.1: Project Areas

Atoll Name	Estimated km ²	Coverage Achieved (km ²)
Nanumea	32.10	23.67
Nanumaga	18.30	5.84
Niutao	6.60	4.29
Nui	28.20	22.24
Vaitupu	16.00	12.97
Nukufetau	156.50	124.16
Funafuti	300.40	245.67
Nukulaelae	61.40	45.76
Niulakita	2.40	5.66

2.2.4 Flight Planning and Operations Description

All Nine Atolls were divided into separate survey Sub-Areas (SA) allowing:

- Efficient data collection,
- Turbidity management and
- Staged delivery across the course of the project.

Figure 2.1 to Figure 2.9 show the sub area design Fugro used for each of Tuvalu's Nine Atolls.

In total, 11 survey flights were conducted. A summary of dates can be found in Annex I.

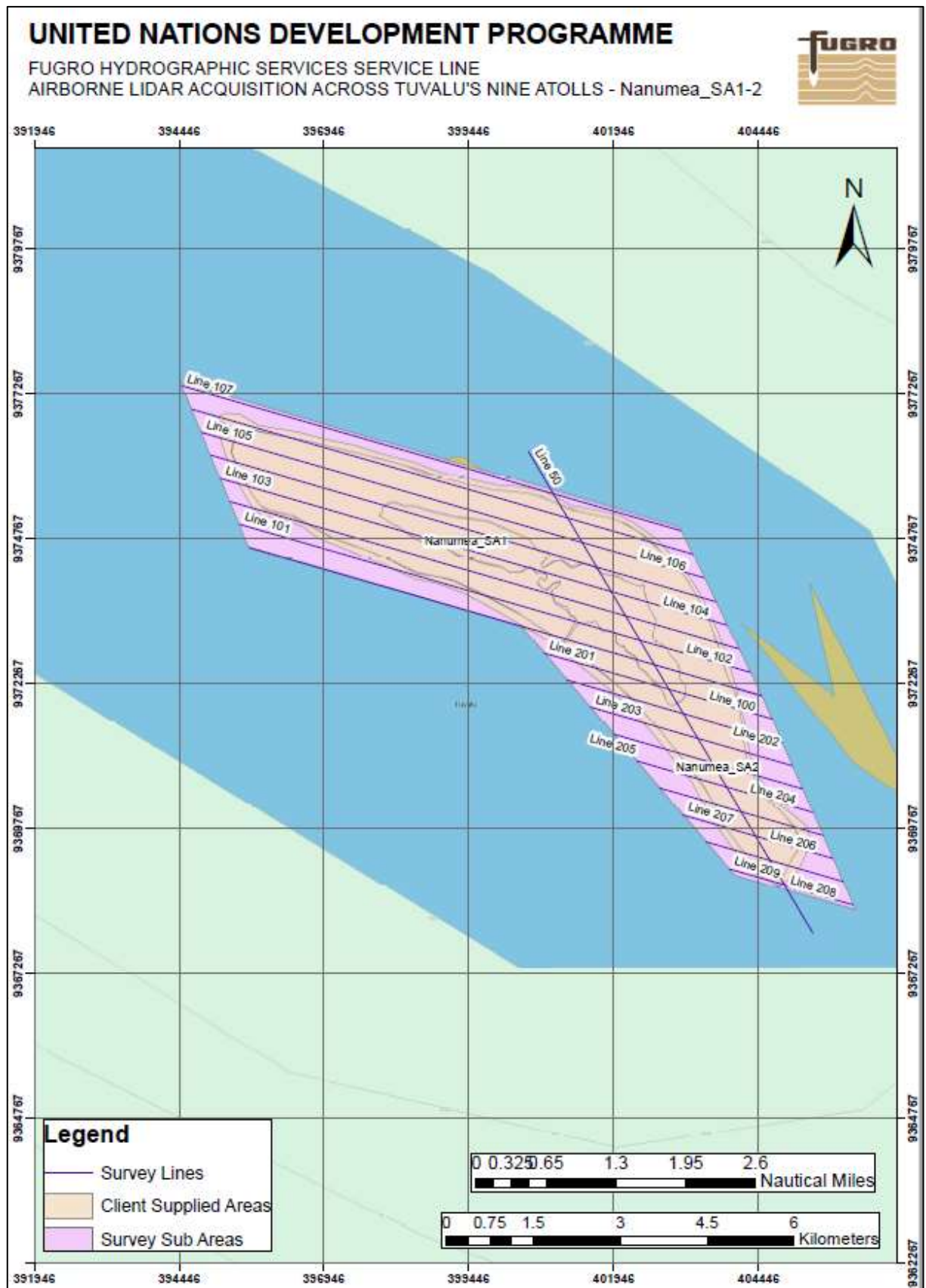


Figure 2.1: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Nanumea

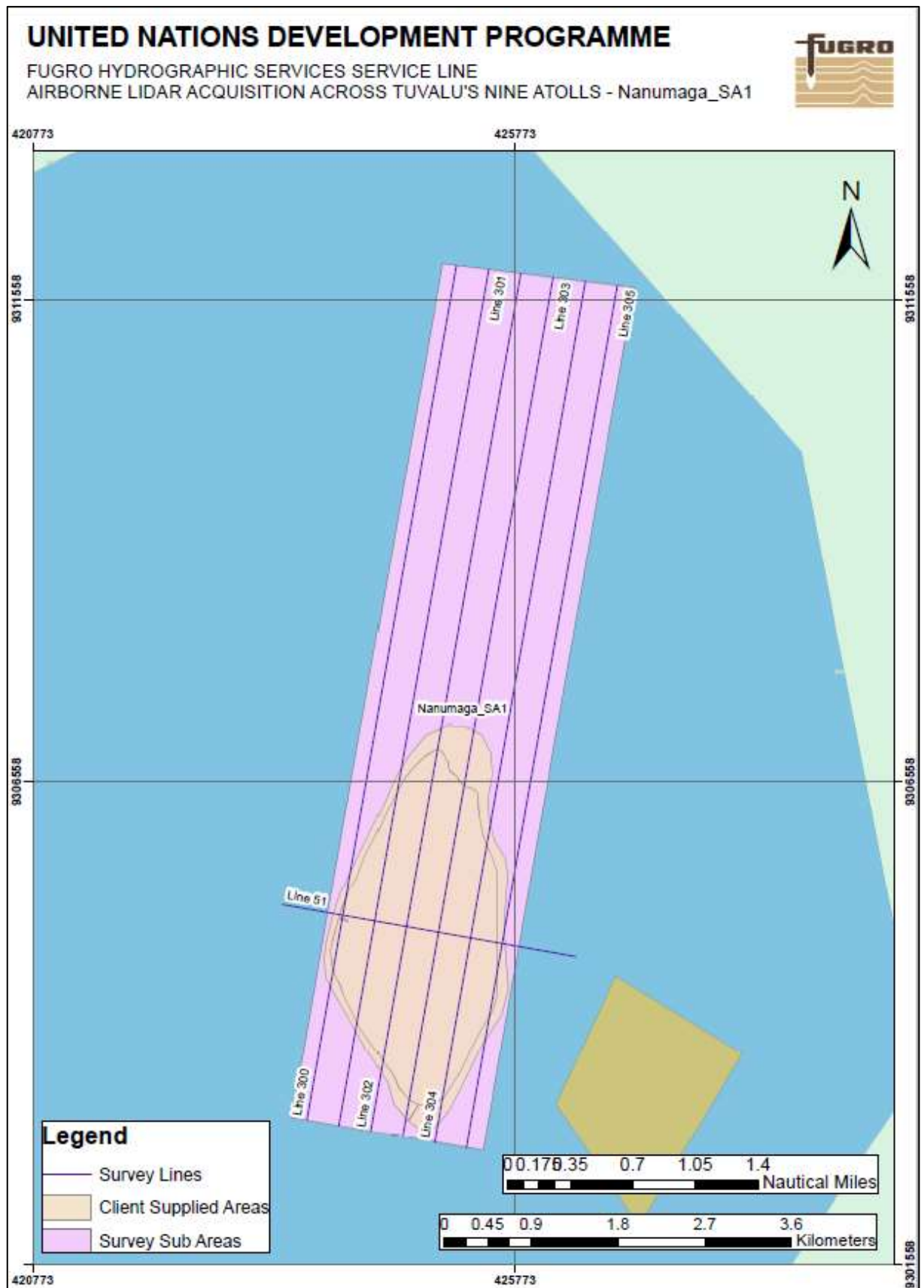


Figure 2.2: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Nanumaga

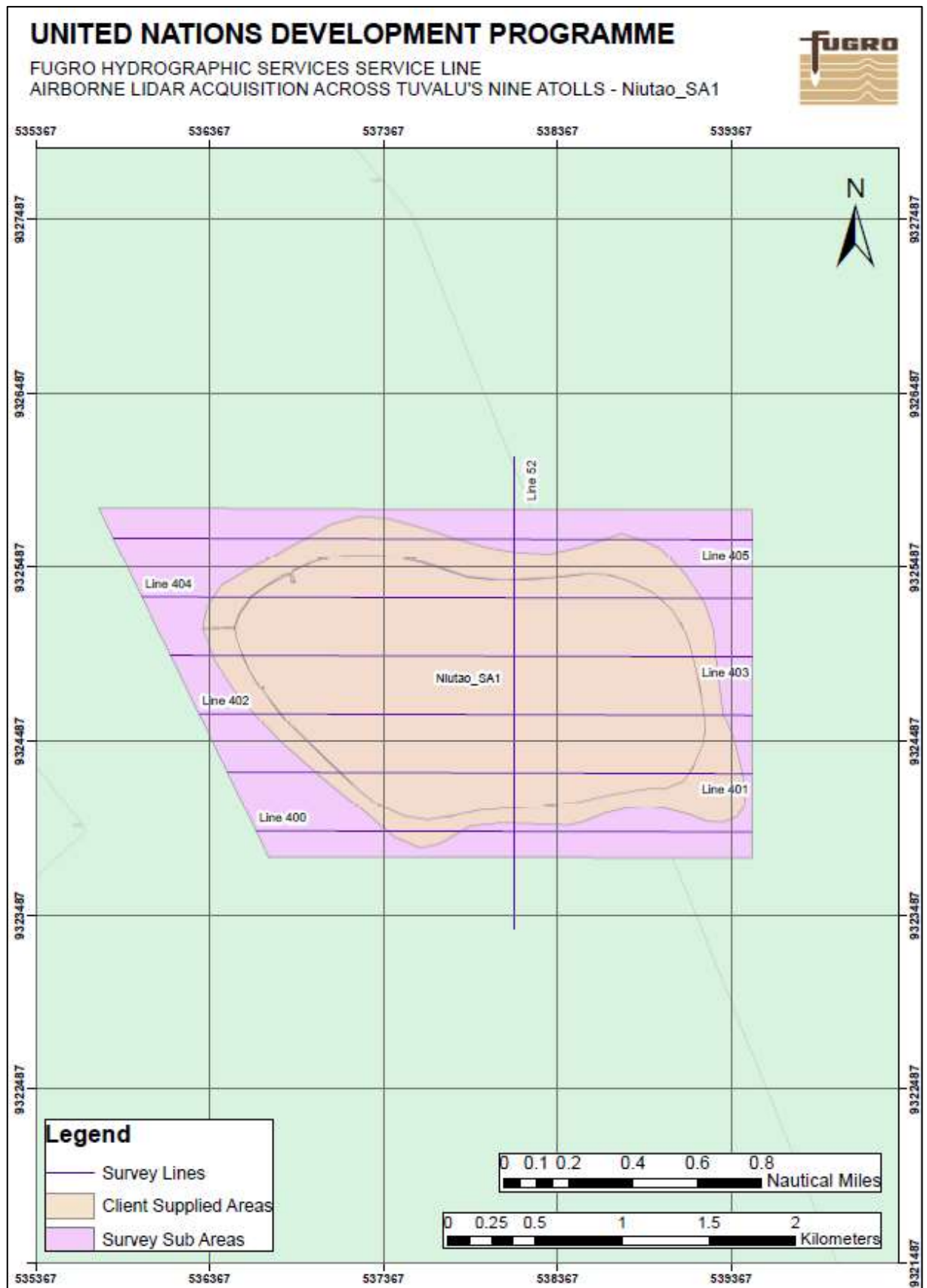


Figure 2.3: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Niutao



Figure 2.4: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Nui

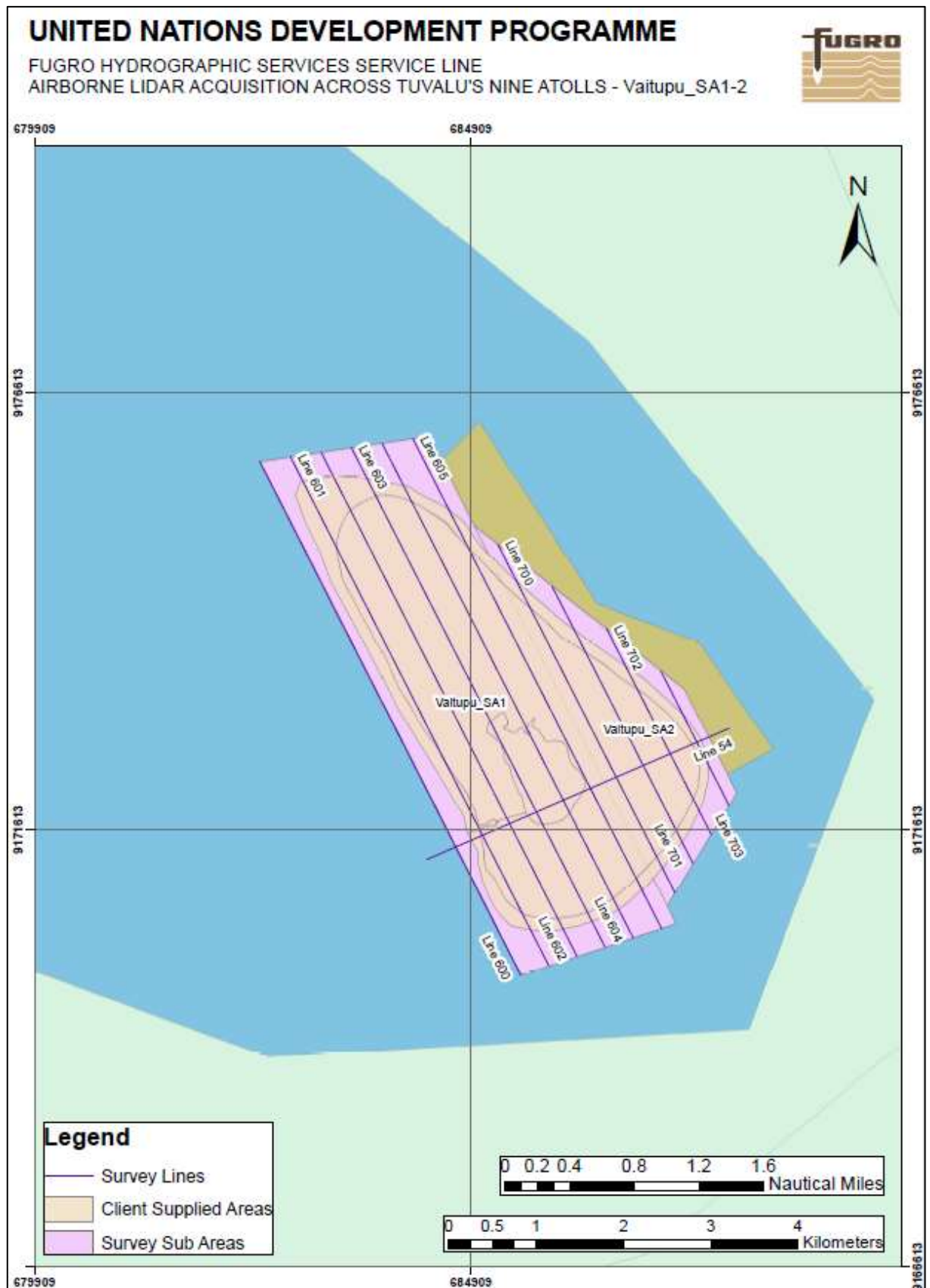


Figure 2.5: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Vaitupu

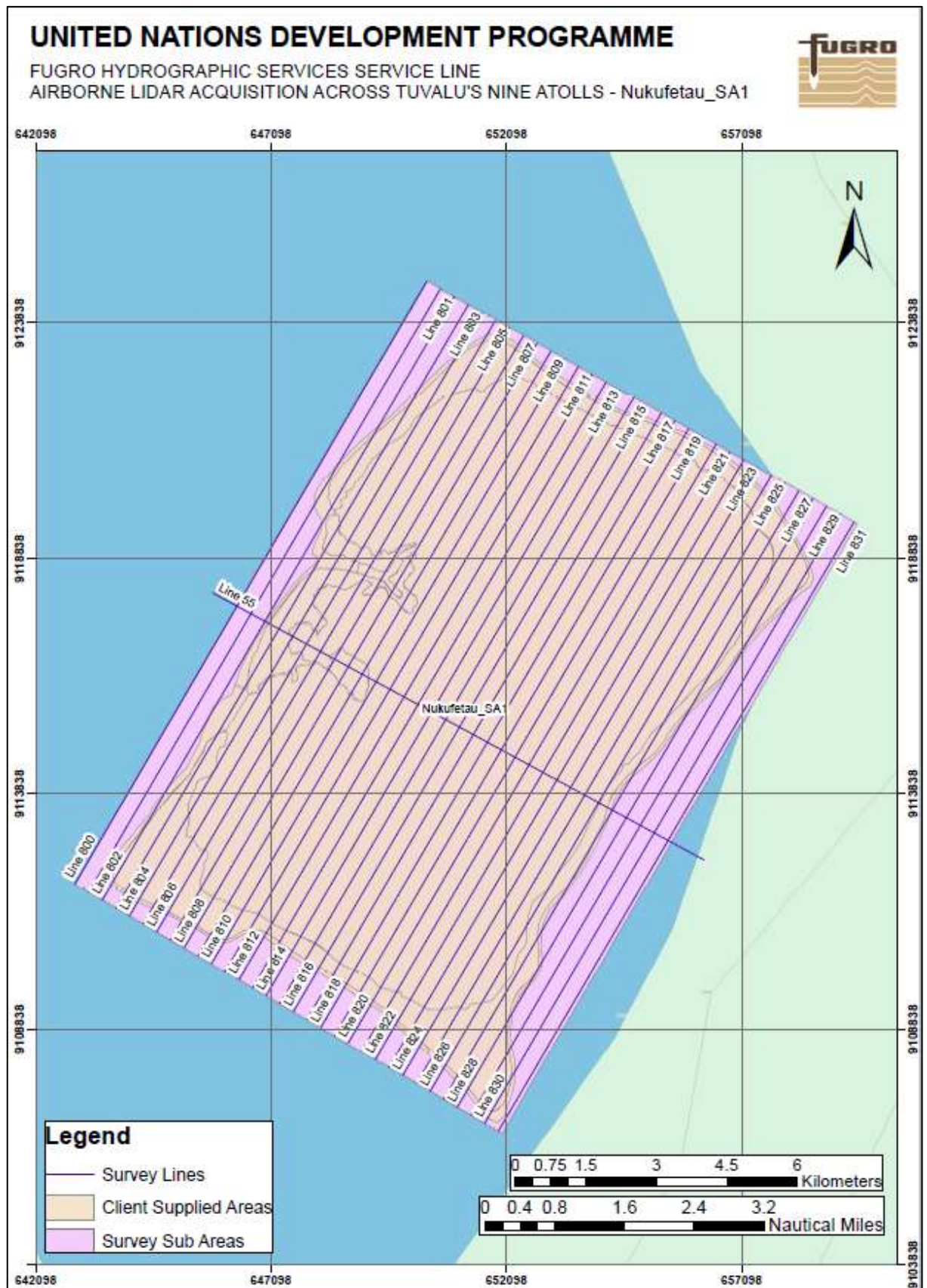


Figure 2.6: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Nukufetau

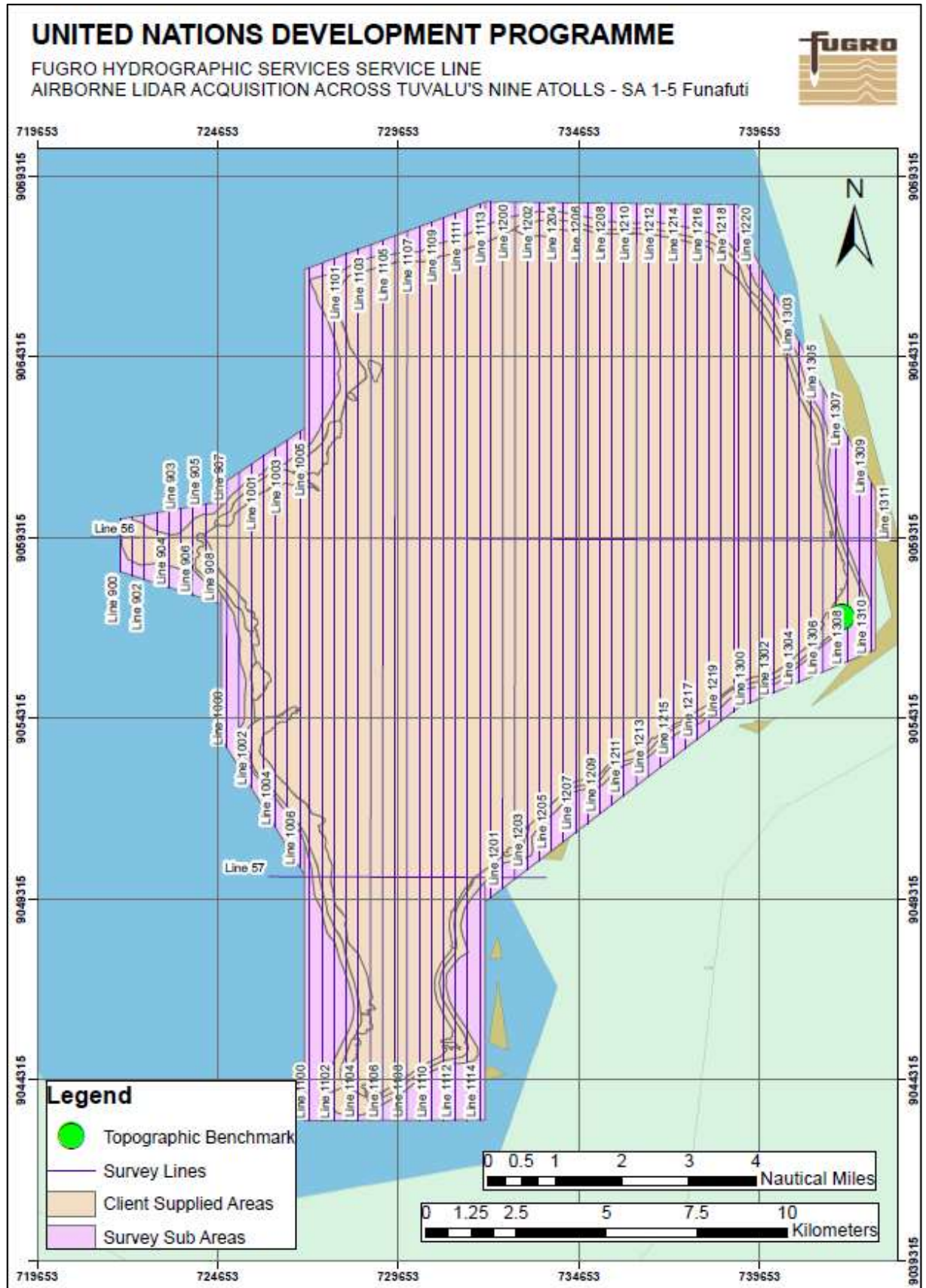


Figure 2.7: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Funafuti

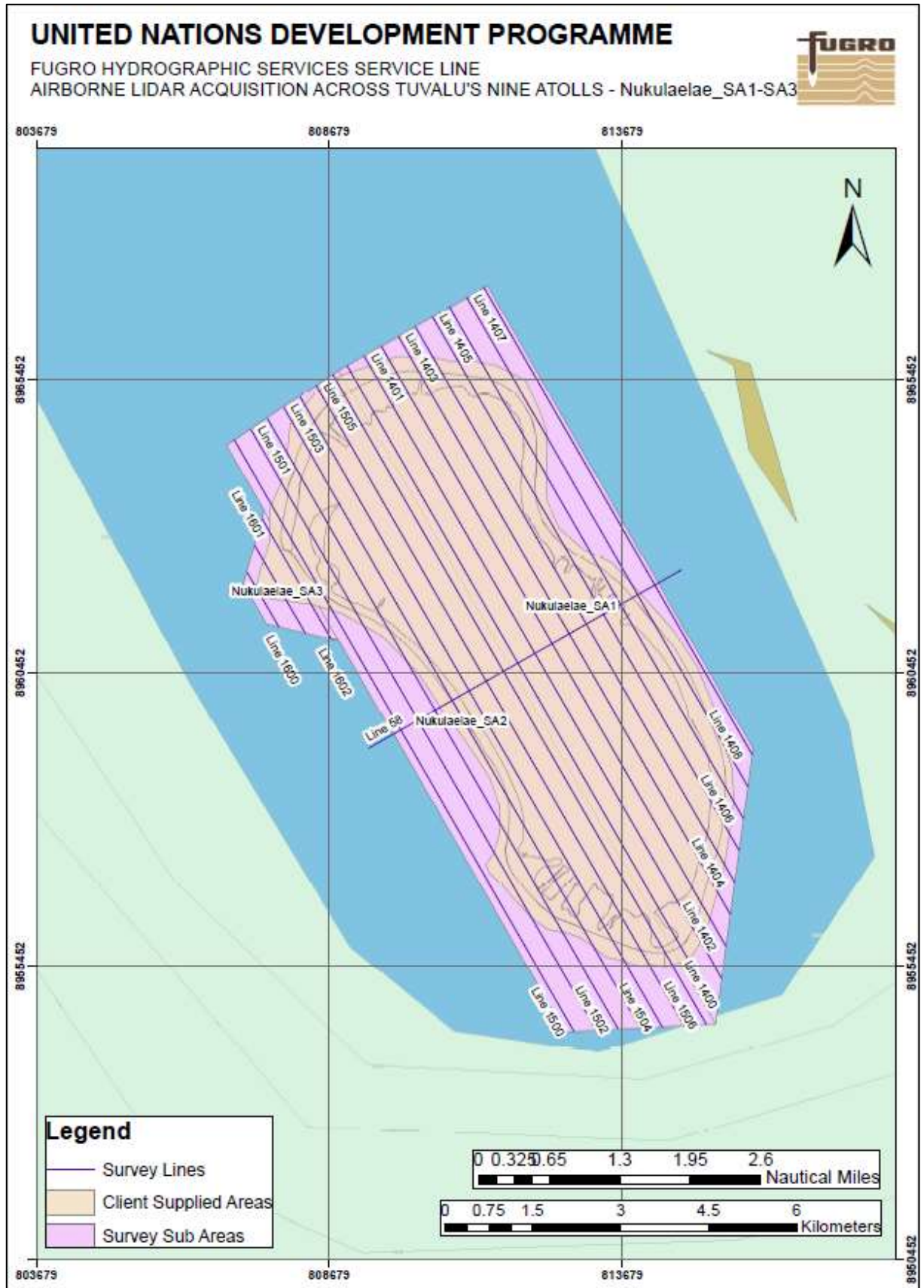


Figure 2.8: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Nukulaelae

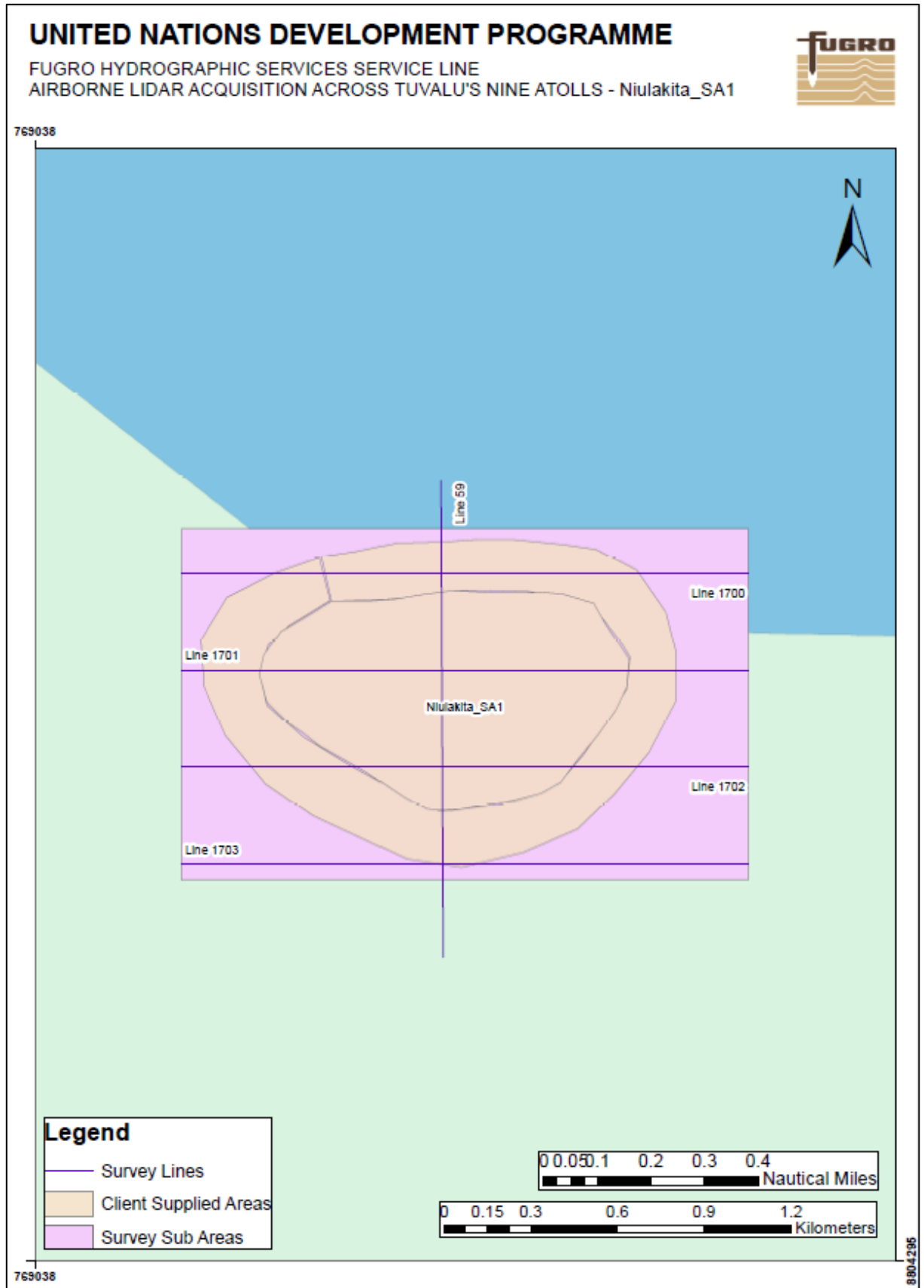


Figure 2.9: LADS HD and RIEGL VQ-820-G Combined Marine LiDAR Flight Planning – Niulakita

2.2.5 Weather and Environmental Conditions

Heavy rainfall and poor conditions were prevalent in the months of April and May 2019, which delayed the start of the survey operations. During the actual survey period, from 24 May through to 3 June 2019, conditions were relatively clear with only 1 day affected by low cloud and rainfall overall.

Survey operations were initially planned for 1800 feet however cloud cover could often form at 1500 feet during the course of the day. To maintain flexibility in daily flight operations survey areas were replanned to incorporate a 1400 feet option and flown for a total of 3 out of the 11 flights conducted.

2.2.6 Daily Survey Logs

Daily Survey Logs were compiled for each day of the deployment and distributed to the Fugro survey and management teams to track progress.

2.2.7 Personnel

The following personnel in Table 2.2 and Table 2.3 were involved in the data collection and processing.

Table 2.2: Field Team Composition

Name	Position	Company
Luke Chamberlain	Party Chief	Fugro
Huw Thomas	Deputy Party Chief	Fugro
Ryan Wilkins	Surveyor	Fugro
Tristan Haynes	Data Processor	Fugro
Raj Sekar	Engineer	Fugro
Matthew Goodall	Pilot	Corporate Air
Angus Robertson	Pilot	Corporate Air

Table 2.3: Processing, Products and Reporting Team Composition

Name	Position	Company
Benoit Cajelot	Project Manager	Fugro
Luke Chamberlain	Party Chief	Fugro
Darren Stephenson	Party Chief	Fugro
Lucie Migne	Data Processing Manager	Fugro
Nic Bailey	Imagery and Products Coordinator	Fugro
Ben McWilliam	Geodetics Technical Lead	Fugro
Tom Gostlow	Products Technical Lead	Fugro
Phuong Le	Surveyor	Fugro
Ryan Wilkins	Surveyor	Fugro
Jarrad Hooper	Surveyor	Fugro
John Bird	Surveyor	Fugro
Huw Thomas	Surveyor	Fugro
Tristan Haynes	Data Processor	Fugro
Sarah White	Data Processor	Fugro
Harry Newsham	Data Processor	Fugro

3. TECHNICAL DESCRIPTION

3.1 Equipment

3.1.1 Aircraft

The LADS HD and RIEGL VQ-820-G ALB systems were installed in a Cessna 441 aircraft operated by Corporate Air. The Cessna 441 has a transit speed of 250 knots at altitudes of up to 26,000 feet and an endurance of up to four hours. Survey operations were conducted at 1800 and 1400 feet at 160 knots.



Figure 3.1: Corporate Air Cessna 441 – VH-VEH

3.1.2 LADS HD LiDAR Surveying System

The LADS HD system uses an Nd:YAG laser on a fixed platform, which operates at 3024 hertz. The electro-mechanical scanner can provide dozens of different laser spot spacing configurations, from 2.0m to 4.8m point density, at altitudes between 1200ft and 3000ft and ground speeds of 130 knots to 175 knots. The scanner compensates for the aircraft's roll and cross track error. The green returned laser energy is captured by the green receiver, and then digitised and logged to a Secure Digital (SD) Card. The height of the aircraft is determined from the green laser, IMU and GNSS. The system can operate during both day and night. Night operations are enhanced by removing the filter on the receiving optics. Real-time positioning is provided by Fugro Marinestar® GNSS PPP or raw GNSS. The POS AV contains an IMU and GNSS logger (roving receiver) that is used in the calculation of post-processed KGNSS positions using POSpac MMS or Inertial Explorer processing software.

The LADS HD laser spot spacing of 3.4m x 3.4m and 3.0m x 3.0m was utilised for this survey. The achieved swath width of the 3.4m scan pattern at 1800ft altitude is 356m. The achieved swath width of

the 3.0m scan pattern at 1400ft altitude is 277m. The spacing of mainlines were planned at 336m for 1800ft lines and 257m for 1400ft lines to provide sufficient overlap between adjacent lines.

The LADS HD airborne system comprised of the following shown in Table 3.1. Figure 3.2 also shows the installation of the LADS HD equipment inside VH-VEH.

Table 3.1: LADS HD Equipment

System	Description	Manufacturer
ALB	Fugro LADS HD (Build 2 – B2)	Fugro
Positioning	BD982 GNSS Receiver	Trimble
IMU	Applanix POS AV 510 IMU	Applanix
Digital Imagery	Phase One IXU-RS-1000 100MP digital camera	Phase One
Interface	ToughBook Linux operator laptop	Dell



Figure 3.2: The Operator and Observer Seats and the LADS HD System Inside the Aircraft

Refer to Annex B for further details on the Fugro LADS HD LiDAR system

3.1.3 RIEGL VQ-820_G LiDAR System

The RIEGL VQ-820-G system uses an Nd:YAG laser on a fixed platform, which operates at 148 kHz, 284 kHz or 512 kHz. For this survey the RIEGL VQ-820-G system (Figure 3.3) was operated at 284 kHz to provide nominally 0.5 m x 0.5 m laser spot spacing across a swath width of ~420 metres at 1800ft and ~330 metres at 1400ft. The scanner is a rotating mirror and so no off-track or roll compensation is applied.



Figure 3.3: The RIEGL VQ-820-G System Installed Behind the Operator's Seat

The green returned laser energy is captured by the green receiver, and then digitised and logged to a hard drive. The system can operate during both day and night. RIEGL's RiAcquire software is used for system setup and data logging. Real-time positioning is provided by logging raw GNSS. The POS AV IMU contains an IMU and GNSS logger (roving receiver) that is used in the calculation of post-processed KGNSS positions using POSpac MMS or Inertial Explorer processing software.

In summary the RIEGL VQ-820-G airborne system comprised of the following equipment itemised in Table 3.2. Refer to Annex C for further details on the RIEGL VQ-820-G system.

Table 3.2: RIEGL VQ-820-G Equipment

System	Description	Manufacturer
ALB	RIEGL VQ-820-G (Serial Number S2220710)	RIEGL
Positioning	BD982 GNSS Receiver	Trimble
IMU	Applanix POS AV 510 IMU	Applanix
Interface	ToughBook Linux operator laptop (same as LADS HD)	Dell

3.1.4 Phase One Imagery Camera

A digital imagery system is mounted on the Fugro LADS HD platform and comprises of a Phase One IXU-RS-1000 100MP digital camera, a Matrox 4sight M frame grabber and a Matrox embedded computer running a Windows XP operating system. Images are taken at three-second intervals with a 11608 x 8708 resolution and a 100-megapixel interline-transfer camera head and controller. At the end of each day, the flight images were resampled to 5 megapixels and then included and imported into the GS for assistance in data validation. The full resolution images were backed up to the field NAS. The Phase One imagery ultimately had quality control and mosaic generation applied to it. Details of this can be found in Annex B.



Figure 3.4: Phase One IXU-RS-1000 100MP Digital Camera

3.2 Geodetic Systems and Horizontal Survey Control

All collected data was post-processed relative to the International Terrestrial Reference Frame 2014 (ITRF2014@2019.4) datum during the application of the Kinematic Global Navigation Satellite System (KGNSS) solution.

Initial data deliveries were produced using UTM60S, ITRF14 with vertical component relative to the GRS80 Ellipsoid. Final data deliveries were produced with vertical component relative to the MSL using undulation models provided by SPC.

GNSS Continuously Operating Reference Stations (CORS) were used for the post processed positioning during the set to work flights prior to deploying to Tuvalu. This service was provided by SmartNet AUS. As no CORS system existed across Tuvalu's Nine Atolls, all survey activities were post processed using a PPP solution.

Additional survey control was established as part of the set to work activities on the 15 May 2019 where Fugro personnel coordinated a Static Control Position Mark at Goulburn Airport from a four-hour logging session. This mark was used for the static position check conducted just prior to the coordination. In order to derive the antenna position for each respective ALB system, the aircraft heading was determined by the Applanix Inertial Measurement Unit (IMU) whilst logging the GNSS data. This allowed for a gross error check to be calculated for both the LADS HD and RIEGL VQ-820-G positioning systems.

Full details on all Set to Work activities, refer to Fugro Document No. TLC00.066.005-TUV 1 Tuvalu_Set To Work & Verification Report_Rev_3.

3.3 Position Fixing Systems

3.3.1 LADS HD and RIEGL VQ-820-G LiDAR Positioning System

Throughout the survey, the real-time position of the LADS HD system was derived from a Trimble BD982 GNSS receiver. Real-time PPP corrections from the Fugro Marinestar® GNSS service were received and applied by the Trimble receiver to the raw GNSS positions. The RIEGL VQ-820-G GNSS receiver was also setup to receive these corrections to assist with the post-processing of the GNSS data however the LADS HD GNSS system is always used for the real-time position of the aircraft.

L1/L2 carrier phase PPK GNSS positions from the Trimble BD982 (POS AV) multi-frequency GNSS receivers on the aircraft were calculated using Inertial Explorer. These positions were imported into the GS and RiProcess software and applied to all soundings. Details of the position fixing systems are provided in Annex E.

3.3.2 Horizontal Accuracy Theoretical Estimates

A theoretical assessment of the total survey accuracy can be determined by combining the errors due to the GNSS positioning systems, the frame centre reference positions from GNSS fixes, platform and laser positioning errors and sea surface errors. The horizontal accuracy of the survey has been assessed theoretically using this method and is presented in Annex H.

The theoretical analysis of the pool of errors indicates that the systems employed, and survey practices resulted in IHO Order 1b standards being achieved or better. This is confirmed by the static position check results described in Section 3.3.3 below.

3.3.3 Practical Horizontal Accuracy Checks

A static position check of the ALB systems was conducted on the apron at Goulburn Airport on 15 May 2019 prior to data collection phase to ensure the positioning systems were operating correctly for the survey. During the static position check, data was logged on board the aircraft for over two hours and the LADS HD laser source was plumbed to the ground. Once the session had finished, the aircraft was moved away from the plumbed point so Fugro personnel could coordinate the point using GNSS logging equipment. After the GNSS data from the aircraft and ground point were processed, the absolute coordinates for the laser source was deduced. Using the aircraft heading and antenna offsets enabled the absolute position of both the LADS HD and RIEGL VQ-820-G antennae to be derived and compared to the real-time position.

Results of the horizontal and vertical static accuracy checks of both the LADS HD and RIEGL VQ-820-G systems are provided in Table 3.3. The results for both ALB systems are consistent with previous surveys and show that the positioning systems were functioning correctly.

Table 3.3: Static Position Accuracy Results for LADS HD and RIEGL VQ-820-G

GNSS Positioning System	LADS HD		RIEGL VQ-820-G	
	Absolute Horizontal Accuracy 2D (m) (95% confidence)	Absolute Vertical Accuracy 1D (m) (95% confidence)	Absolute Horizontal Accuracy 2D (m) (95% confidence)	Absolute Vertical Accuracy 1D (m) (95% confidence)
WADGNSS	0.169 m	0.306 m	1.238 m	3.394 m
KGNSS PPP	0.068 m	0.125 m	0.154 m	0.101 m
KGNSS BS	0.059 m	0.128 m	0.057 m	0.110 m

Details of LiDAR position checks are provided in Annex E

3.3.4 Accuracy of Position

The initial checks on positional accuracy for both LiDAR systems were presented in the "TLC00.066.005-TUV 1 Tuvalu_Set To Work & Verification Report_Rev_3", extracts of which are included in Annex E.

Based on the results from the static position checks it is considered that the minimum IHO Order 1b positional accuracy or better was achieved (5m at 95% confidence level).

Full details of the LiDAR position accuracy are provided in Annex H.

3.4 Vertical Datum and Reduction

3.4.1 Survey Datum

All soundings for the initial data collection have been reduced directly to the Geodetic Reference System 1980 (GRS80) ellipsoid and simultaneously to the real-time sea surface datum.

Final data was supplied relative to Mean Sea Level (MSL) at Funafuti Tide Gauge, defined as 2.048m above chart datum. This was reduced using a GEOID undulation model for each atoll provided by SPC or adopting the EGM08 global geoid model with a local mean sea level correction.

3.4.2 Tide Model

No tidal model was used in the reduction of the Tuvalu survey.

Geoid models relating the GRS80 ellipsoid to MSL were derived for each atoll. Refer to Annex F for more details on the specific methods adopted for each atoll.

3.5 Vertical Accuracy

3.5.1 Vertical Accuracy Theoretical Estimates

A theoretical assessment of the total survey accuracy can be determined by combining the errors due to the LADS HD and RIEGL VQ-820-G ALB systems, the vertical datum (i.e. tides and tidal model or Geoid model), swell and water clarity. The depth accuracy of the survey has been assessed theoretically using this method and is presented in Annex H.

The theoretical analysis of the pool of errors indicates that the systems employed, and survey practices resulted in IHO Order 1b standards being achieved or better. This is confirmed by the depth and TIP benchmarks and cross line comparison results described in Sections 3.5.2 to 3.5.4 below.

3.5.2 Depth Benchmarks

Unfortunately, no depth benchmark data was available in Tuvalu during the course of the survey. **Update with historic 2004 MBES data from SPC when received.**

Bathymetric vertical accuracy was verified prior to survey data during the set to work period in Wollongong. The location of these benchmarks, in the vicinity of Wollongong, are shown in Figure 3.5 and Figure 3.6 below and the summary of the results are presented in showing good agreement.

No RIEGL VQ-820-G data overlapped any historic or new bathymetric benchmarks that were shallow enough during the set to work. However, results from the topographic benchmark comparisons, outlined in section 3.5.3, and comparisons with the LADS HD bathymetric data outlined in Section 3.6 demonstrate that the RIEGL VQ-820-G is fit for purpose.

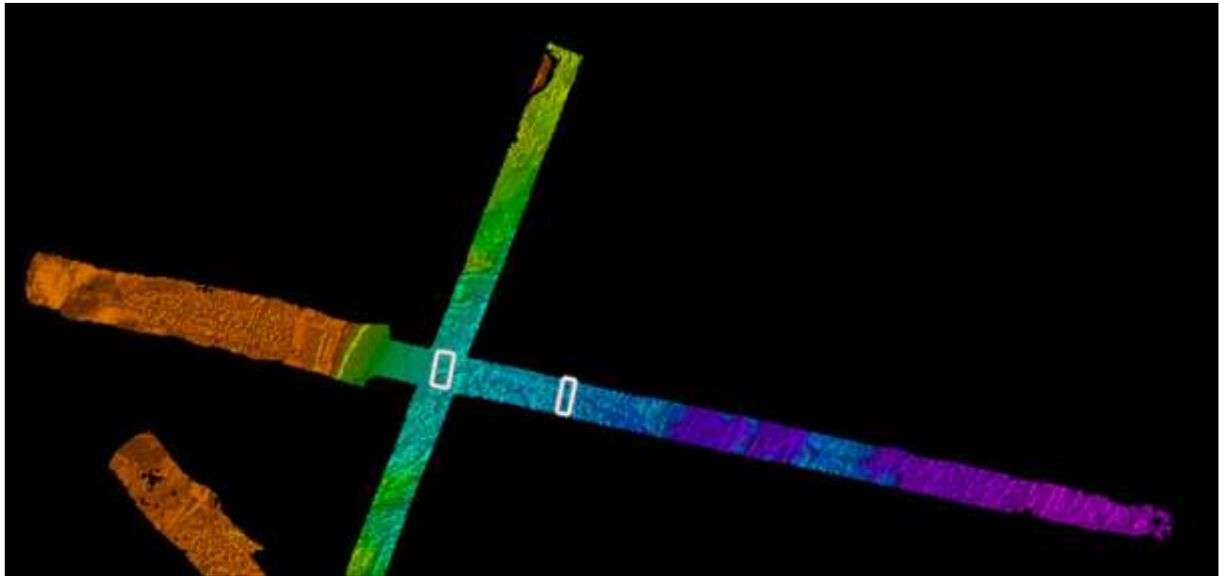


Figure 3.5: Wollongong MBES Benchmarks – BM 27 (West, 21m Average AHD Depth) and BM 28 (East, 28.5m Average AHD Depth)

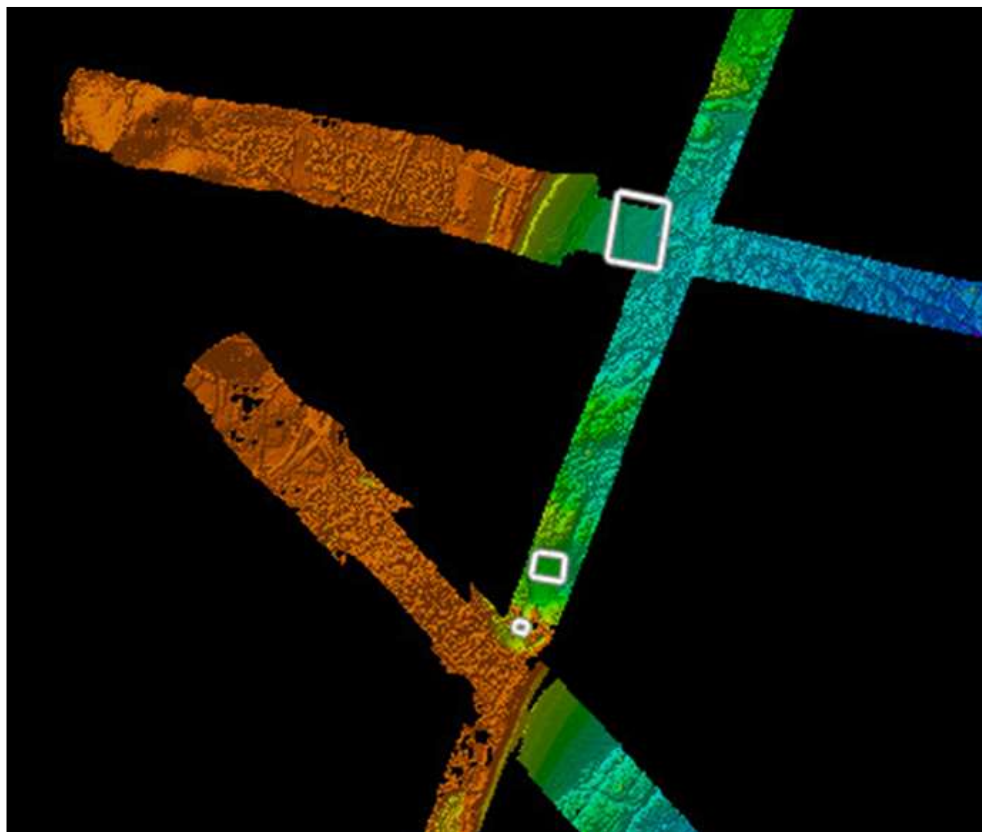


Figure 3.6: Wollongong ALB Benchmarks – BM 22 (North, Mean Depth 17.3m Below MSL), BM 35 (South, Mean Depth 12.5m Below MSL) and BM 36 (Centre, Mean Depth 9.1m Below MSL)

Table 3.4: LADS HD Benchmark Results Summary Against MBES Data & Historical ALB Data

Pattern	No. of Comparisons	Mean Depth Difference (m)	Standard Deviation (m)
H34D18F356	91465	-0.03	0.17

For further details of the depth benchmarks and individual BM results, refer to Annex G and Fugro Document TLC00.066.005-TUV 1 Tuvalu_Set To Work & Verification Report_Rev_3.

3.5.3 Topographic Benchmarks

Topographic Integration Point (TIP) comparisons were conducted over two newly established locations on Funafuti. These were used to verify the accuracy and precision of the LADS HD and RIEGL VQ-820-G systems in the topographic environment. The Funafuti TIP's were relatively flat and independently surveyed by Fugro and Tuvalu's Land and Survey Department personnel using Real Time Kinematic (RTK) GNSS land survey techniques. The TIP was surveyed in ITRF2014, UTM 60S with heights relative to the GRS80 ellipsoid.

Both TIPS were located near the Funafuti International Airport and flown 5 times during survey operation flights at 1800ft and 1400ft altitudes. A summary of the combined TIP comparisons for the LADS HD and RIEGL VQ-820-G are shown in Table 3.5. These results are comparable with previous results obtained with the LADS HD and RIEGL VQ-820-G systems and show they operated correctly.

Table 3.5 ALB Sensor TIP Results Summary for Combined TIP1 and TIP2

ALB System	Number of Comparisons	Ellipsoid	
		Average MDD	Average SD (1 σ)
LADS HD	8911	-0.08	± 0.06
RIEGL VQ-820-G	564822	0.00	± 0.03

Full results of all final TIP comparisons are provided in Annex G.

3.5.4 Cross Line Comparisons

A total of 11 cross lines were flown, one across each Atoll at least with exception of Funafuti having two. Comparisons on the ellipsoid datum were made with the main survey lines for analysis.

The averages of the mean depth differences and standard deviation for each LADS HD and RIEGL VQ-820-G cross line flown are in Table 3.6. These results are comparable with previous results obtained with the LADS HD and RIEGL VQ-820-G systems and show they operated correctly. Note that the higher standard deviation is attributable to the nature of the seabed with steep slopes in majority of atolls.

Table 3.6: ALB Sensor Crossline Results Summary for Tuvalu's Nine Atolls

ALB System	Number of Comparisons	Ellipsoid	
		Average MDD	Average SD (1 σ)
LADS HD	1587178	-0.01	± 0.50
RIEGL VQ-820-G	55842590	0.01	± 0.32

Full results of the cross-line comparisons are provided in Annex G.

3.6 Merged Dataset Verification

In general the merged overlap between the LADS HD and RIEGL VQ-820-G sensors were between depth range of 5 m to 10 m (or extinction depth of the RIEGL VQ-820-G). As a check on the sensor

agreement the overlap points in Funafuti, being the largest atoll in Tuvalu, was calculated and a summary showing good agreement is shown in Table 3.7.

Table 3.7 LADS HD and RIEGL VQ-820-G Merged Comparison Results - Funafuti

Summary of the Combined Sensor Overlap (m)				
Atoll	Number of Comparisons	Average Difference	Standard Deviation	Comments
Funafuti	3024255	0.14	0.35	Riegl below LADS HD

See Annex G for more information.

3.7 Phase One Horizontal Position Accuracy

Assessment of the positional accuracy of the Phase One digital camera was carried out against ground check points in Funafuti Runway, Tuvalu. These ground controls points (GCP's) were surveyed by Fugro and Tuvalu's Land and Survey Department personnel using Real Time Kinematic (RTK) GNSS land survey techniques, reported in Annex D.

The Phase One calibration data was processed using POSPac MMS Photogrammetry Tools, in particularly the calibration and quality control software package. The GCP and check point results are displayed in Table 3.8.

Table 3.8: Phase One Ground Control Points Result Summary

ITRF14 UTM60S	Surveyed Check Points		Difference Summary	
	Easting	Northing	Easting Difference	Northing Difference
1	741693.17	9056968.73	-0.07	-0.13
2	741695.27	9056966.67	-0.08	0.07
3	741710.62	9056982.95	0.28	-0.01
4	741708.57	9056985.02	-0.08	-0.08
5	741723.17	9056970.58	-0.13	-0.07
6	741725.37	9056968.45	-0.08	0.01
7	741710.03	9056952.06	-0.09	0.11
8	741707.93	9056954.20	-0.07	0.00
9	741929.62	9057188.54	-0.08	-0.01
10	741931.81	9057186.45	-0.19	0.10
11	741947.07	9057202.83	-0.13	0.08
12	741944.94	9057204.87	-0.09	0.02
13	741931.98	9057217.15	-0.27	0.00
14	741929.75	9057219.28	-0.27	0.03
15	741914.43	9057203.00	-0.12	0.15
16	741916.63	9057200.87	-0.07	0.17
17	742032.31	9057296.98	-0.19	-0.04
18	742034.45	9057294.95	-0.25	-0.01
19	742049.83	9057311.22	-0.22	0.02
20	742047.63	9057313.30	-0.19	-0.05
21	742034.49	9057325.48	-0.16	0.05
22	742032.24	9057327.59	-0.23	0.09
23	742016.80	9057311.30	-0.04	0.25
24	742019.06	9057309.14	0.01	0.24
	Average		-0.12	0.04

The results of the independent GCP check in Table 3.8 demonstrate that the Phase One products meet the desired quoted horizontal accuracy of IHO Order 1b or better.

For a further detailed analysis of the Phase One Horizontal Accuracy checks, refer to Annex G and Fugro Document TLC00.066.005-TUV 1 Tuvalu_Set To Work & Verification Report_Rev_3.

3.8 Summary of Calibration and Verification Results

Table 3.9 presents a summary of the calibration and verification results, detailed in the separate Set to Work and Verification Report titled "TLC00.066.005-TUV 1 Tuvalu_Set To Work & Verification Report_Rev_3". All thresholds are equal to or better than the minimum IHO Order 1b accuracy requirements for the survey.

Analysing all theoretical and practical accuracy checks, it has been concluded that during the survey, IHO Order-1b precision for depth and position was achieved for both the LADS HD and the RIEGL VQ-820-G systems.

Table 3.9: LADS HD and RIEGL VQ-820-G Quality Checks Results Summary

Quality Check	Threshold	LADS HD		RIEGL VQ-820-G	
		Result	Pass/Fail	Result	Pass/Fail
Static Position Check – PPP KGNSS	2m horizontal 0.5m vertical	0.07m horizontal 0.13m vertical	Pass	0.15m horizontal 0.10m vertical	Pass
Static Position Check – BS KGNSS	2m horizontal 0.5m vertical	0.06m horizontal 0.13m vertical	Pass	0.06m horizontal 0.011m vertical	Pass
Horizontal Position Accuracy	2m	0.36m Easting 0.24m Northing	Pass	-0.01m Easting 0.01m Northing	Pass
Topographic Vertical Accuracy (GRS80)	0.5m	0.06m	Pass	0.09m	Pass
Bathymetric Vertical Accuracy (GRS80)	0.5m	-0.03m	Pass	N/A	
Phase One IXU-RS-1000 camera Horizontal Accuracy	0.25m	0.02m Easting 0.14m Northing	Pass	N/A	
Quality Check	Threshold	Merged Dataset			
		Result		Pass/Fail	
Merged Dataset Topographic Verification	0.2m	-0.05m		Pass	
Merged Dataset Bathymetric Verification	0.2m	0.08m		Pass	

3.9 ALB Data Processing Methodology

The ALB processing involved the following stages, outlined in Figure 3.7:

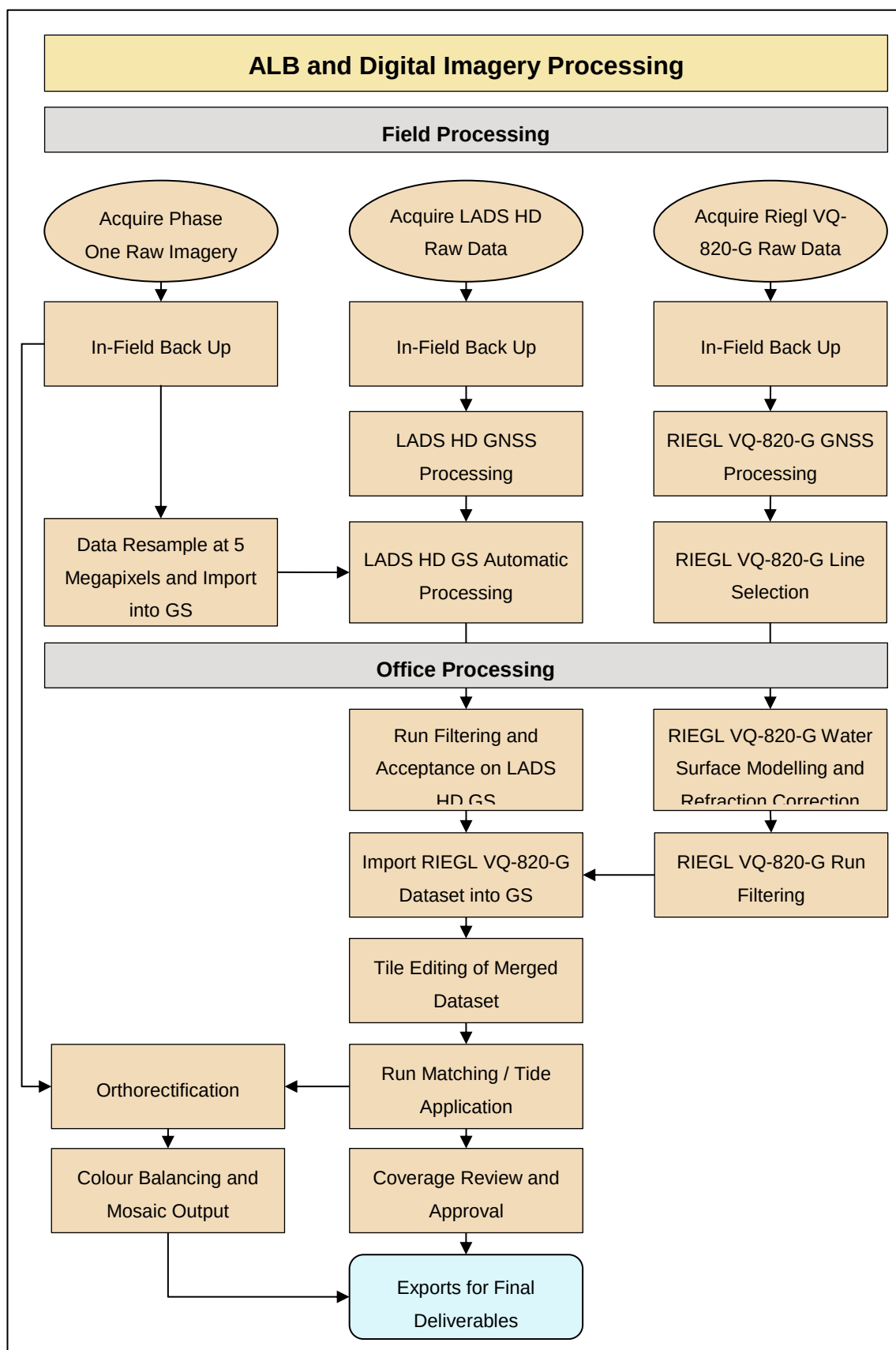


Figure 3.7: ALB and Digital Imagery Processing Flow Diagram

To improve data collection efficiency, all ALB data was copied to a NAS and processed in field allowing for daily coverage assessment. The following processing steps were undertaken:

- Post Processed Kinematic GNSS solution for both sensors using PPP.
- Phase One data resampled from 100 megapixels to 5 megapixels for inclusion into the GS.
- Daily - Backups of all data and transfer of core ALB data and resampled 5 megapixel imagery to external 4TB HD
- Automatic processing and initial batch and filter cleaning of the LADS HD survey run data, which consists of:
 - Examining the depth profile for the correct processing of the survey run.
 - Examining a range of position, coverage and system performance confidences to ensure only good data is accepted.
 - Resolving anomalous soundings by examining data points in various displays.
 - Editing operations which include selection of alternate depths or deletion of soundings as appropriate.
- Initial processing of the RIEGL VQ-820-G data, visualisation and acceptance of runs containing only good data.
- Automatic classification tools were run on the RIEGL VQ-820-G data to remove isolated noise points and define the majority of water surface points where possible.

The ALB data was then processed in the Adelaide office, where the following steps were performed:

- Further refinement of the water surface point selection in the RIEGL VQ-820-G data, creation of a Water Surface Model (WSM) which was then used to correct the bathymetric soundings for the effects of refraction.
- Further refinement of the LADS-HD run based cleaning.
- Merging of RIEGL VQ-820-G and LADS HD data in the same software package and processing of the merged dataset following these area-based processing steps:
 - Tile Validation, which consists of:
 - Datum checks
 - Removal of erroneous data
 - Data re-classification
 - Assessment of ALB extinction depth
 - Tile Checking, which consists of:
 - Review of tile validation
 - Higher level 'QC' of data
 - Generally single point editing
 - Tile Approval, which consists of:
 - Review of tile checking
 - Compilation of Hydro Notes etc.
 - Final approval of dataset

Figure 3.7 shows the individual line processing stages for both the LADS HD and RIEGL VQ-820-G data and merging process, detailed stages of the line processing for each respective system can be found in Annex B and Annex C.

Focus was placed on completing UNDP's priority atolls where possible, and on approval of each area, data was exported out of the GS ground system in the ITRF2014 GRS80 UTM 60 South datum. Products were then created in accordance with Figure 3.8.

3.9.1 Final Point Cloud Classification Table

The following Table 3.10 outlines the point cloud classification classes that have been adopted for all final merged data. Essentially for all "accepted" data that represent the true ground or seabed should include Class 2 (Ground) and Class 13 (Seabed). Detections on vegetation, buildings and other built structures remain as unclassified in class 1.

Table 3.10: Final Point Cloud Classification

Number	Point Class	Description
0	Unclassified	Created, never classified
1	Default	Any non-ground features, such as vegetation and buildings
2	Ground	Ground hits - includes solid breakwaters where water doesn't flow underneath
7	Low/High Points	Spurious high / low point returns, birds, clouds, etc. (unusable)
9	Water Surface	Sea surface points for reference
10	Bridge	Any bridge or overpass over the seabed
13	Bathymetry - Underwater Seabed	Accepted seabed / rock / shoal returns considered IHO Order 1b compliant
14	Bathymetry - Underwater Non-Seabed	Non-seabed and undefined points in water column e.g. turbidity, kelp, sea grass, fish, etc.
16	Fixed Cultural Features	Fixed cultural features of significant interest for navigation such as jetties, wharves, etc.
18	Navigation Aids	Navigation Aids e.g. beacons, buoys, lighthouses
19	Shipwrecks	Shipwrecks
20	Submerged Objects	Submerged features such as pipelines

3.9.2 ALB Processing Software

Table 3.11 outlines the ALB software and version numbers used during the Tuvalu project.

Table 3.11: ALB Processing Software

Software Name	Version	Main Functions
Terramodel	10.61H	CAD software used to create subareas and flight lines during survey planning
Inertial Explorer	8.80.2720	Post processing of the aircraft trajectory using either a base station or PPP
Ground System (GS)	C.3.8.3	The GS offers a wide range of utilities from initial mission planning and imagery georectification through to processing of soundings position and depth, data visualisation, validation, QC and export of digital data
RIPROCESS	1.8.5	RIEGL VQ-820-G raw data processing, visualisation and export to digital format
SDFCopy	1.4	Transfer data from the DR-680 drive. Part of the RiPROCESS suite
POFImport	1.7.4	Convert trajectory files from text to POF. Part of the RiPROCESS suite

Software Name	Version	Main Functions
SDCImport	2.3.7	Extract raw soundings. Part of the RiPROCESS suite
GeoSysManager	2.1.0	Contains the library of Coordinates Reference Systems. Part of the RiPROCESS suite
RIWORLD	5.1.4	Georeference the RIEGL VQ-820-G soundings. Part of the RiPROCESS suite
RIHYDRO	1.8.5	Apply refraction correction to bathymetric soundings. Part of the RiPROCESS suite
TerraScan (application running through Bentley MicroStation)	019.001	LiDAR point cloud visualisation and classification
GS Tile Editor	C.3.8.3	Utility linked to the GS software, used to simultaneously visualise, classify and QC LADS HS and RIEGL VQ-820-G data
Global Mapper	15.2	Visualisation of digital data and coverage plots. Generation of deliverables
ArcMap	10.6	Rendering of coverage plots and contours Generation of deliverables

3.10 Digital Imagery Processing Methodology

The Phase One processing steps were as follows:

- Preliminary coverage and image quality review.
- Resampling to 5cm GSD
- Creation of an external orientation file according to the latest camera calibration specifications and calibration flight
- Pre-colour balance the raw images
- Creation of a digital terrain model
- Orthorectify the accepted runs
- Running a final colour balance and creation of mosaic tiles
- Final QC and product creation

Products were exported as 1km x 1km mosaic tiles in ITRF2014, UTM60S. The final colour balanced mosaic was also used in applying the RGB values to the ALB LAS products.

Table 3.12 outlines the Digital Imagery software and version numbers used during the project. All software used had been upgraded and tested extensively for suitability before the start of the project. No software updates occurred during the Tuvalu survey.

Table 3.12: Digital Imagery Processing Software

Software Name	Version	Main Functions
ImageQC	1.0	Preliminary image and coverage review
Capture One	11.0.0	Pre-colour balancing of Phase One IXU-RS-1000 images
FMOrthoPhoto	2015.2.25	Orthorectification of Phase One IXU-RS-1000 images
OrthoVista	8.0.6	Final colour balancing and creation of mosaic tiles

3.11 Final Deliverables Methodology

Figure 3.8 outlines the workflow followed for the creation of products for the spheroid datum. Figure 3.9 outlines the workflow followed for the creation of products for the MSL datum.

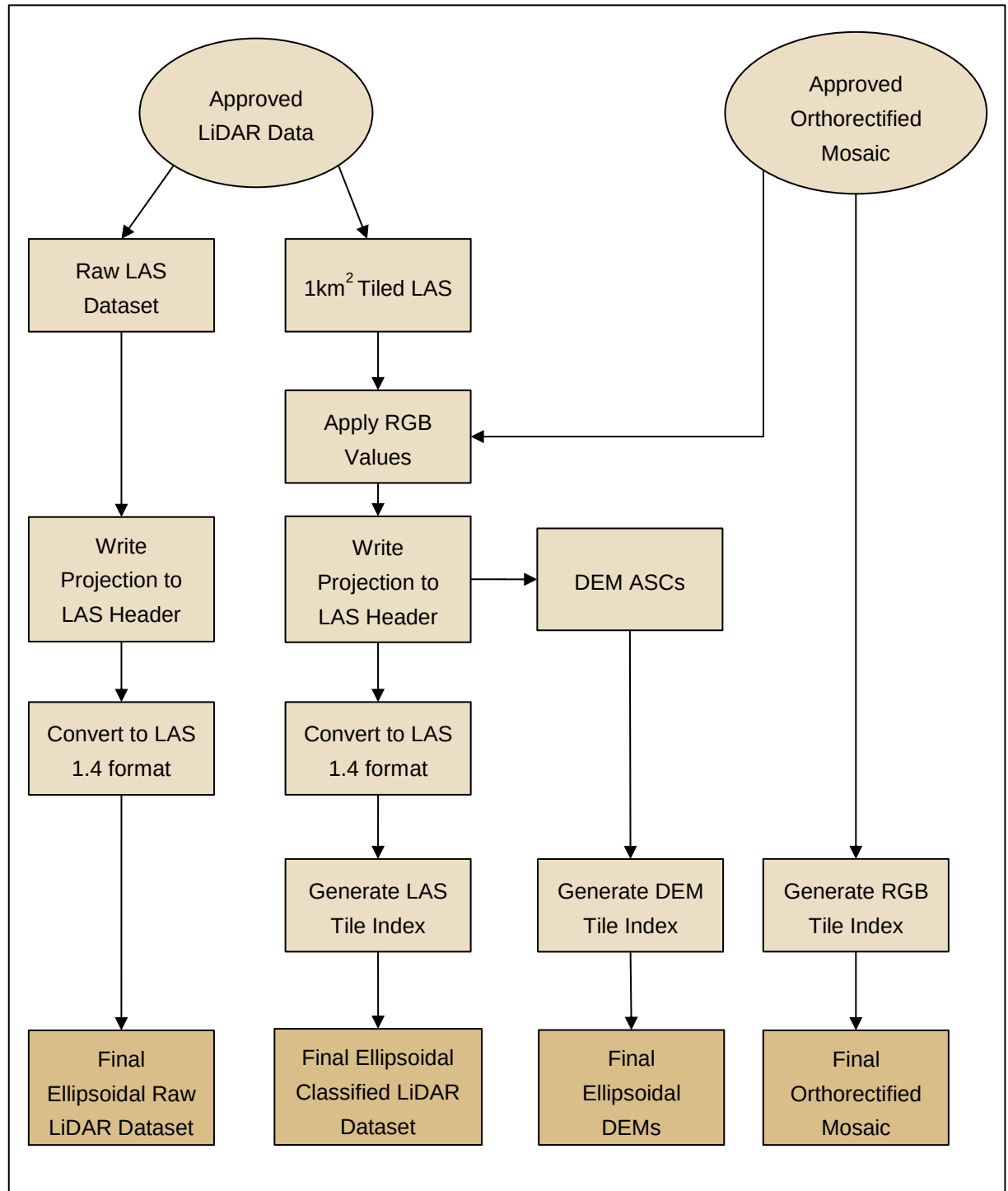


Figure 3.8: Final Ellipsoid Deliverables Workflow

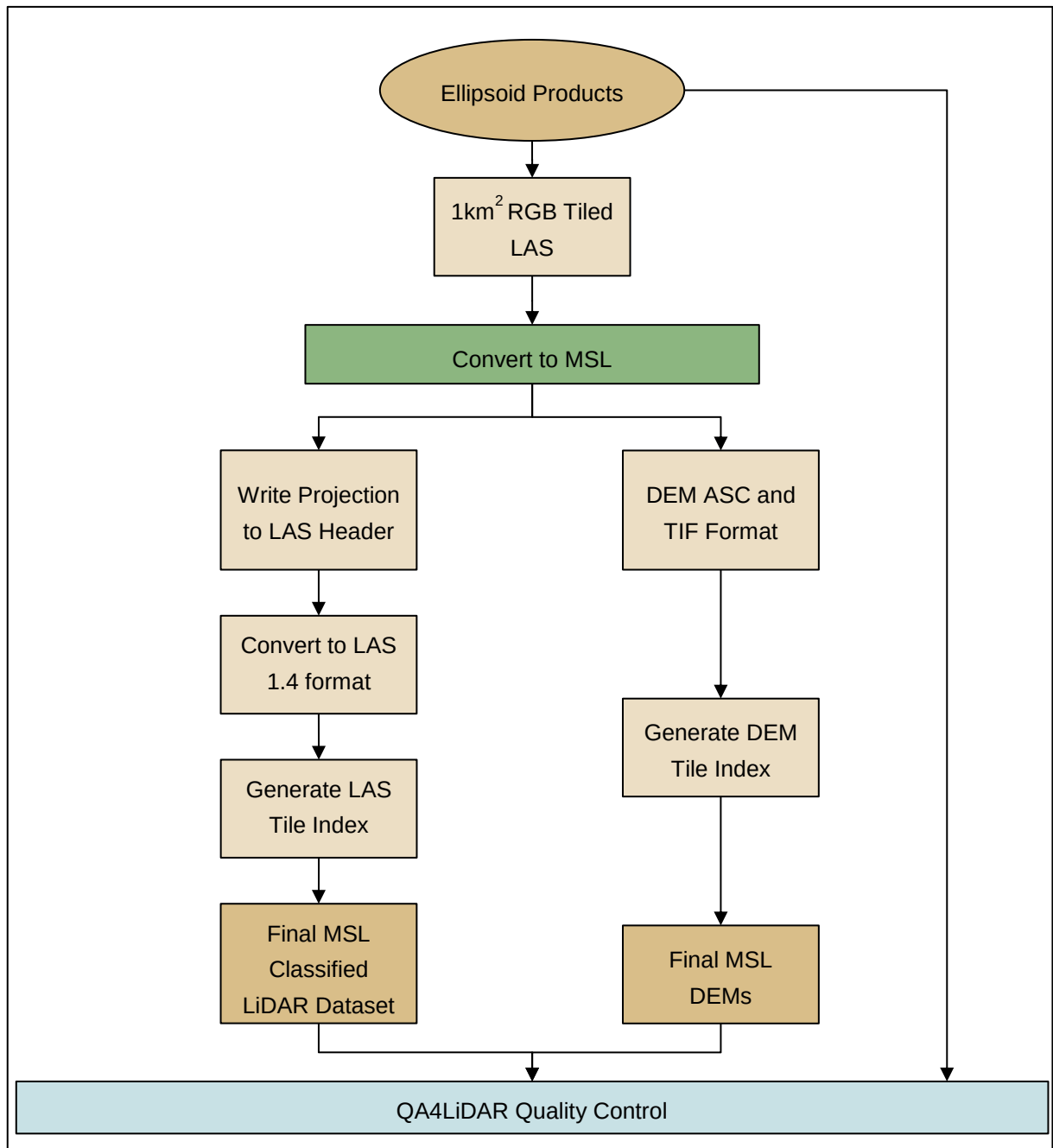


Figure 3.9: Final MSL Deliverables Workflow

Table 3.13 outlines the deliverables software and version numbers used during the project. All software used had been upgraded and tested extensively for suitability before the start of the project. No software updates occurred during the Tuvalu survey.

Table 3.13: Deliverables Processing Software

Software Name	Version	Main Functions
Global Mapper	15.2	Visualisation of digital data and coverage plots. Generation of deliverables
ArcMap	10.6	Rendering of coverage plots and contours Generation of deliverables
LasTools	190507	Generation of deliverables

4. SURVEY RESULTS

4.1 Overview of data collection achieved across Tuvalu's Nine Atolls

Overall excellent results were achieved in the majority of Tuvalu's Nine Atolls project. While the LADS HD system also collects topographic and shallow water data, the denser coverage from the RIEGL VQ-820-G had priority in these areas. As a result, the LADS HD data less than 5m in general was rejected if coverage existed by the RIEGL VQ-820-G system and a 5m depth overlap was maintained generally between the two systems for a seamless dataset. Further descriptions can be found in sections 4.2 to 4.10.

In addition to the nine atolls, seven investigation lines were also flown over a series of sea mounts near Niulakita. Water conditions in this region were very clear and favourable proving the feasibility of further work in this region should the opportunity arise. This is described further in Section 4.11.

4.2 Nanumea

4.2.1 ALB Coverage

Nanumea is the most North-Western Atoll of Tuvalu and made up of the remains of calcareous reef-forming organisms encircling a central lagoon.

Extending from the high waterline, generally out to 300m or more, a shallow reef platform comprised of rocky outcrops and hard coral flats encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over a thousand meters.

Nanumea has two distinct lagoons separated by a series of narrow shallow channels at the centre of the atoll. There is only one access point for surface vessels into the south-east lagoon through a narrow shallow channel along the mid-south approach to the atoll, as shown in Figure 4.1.

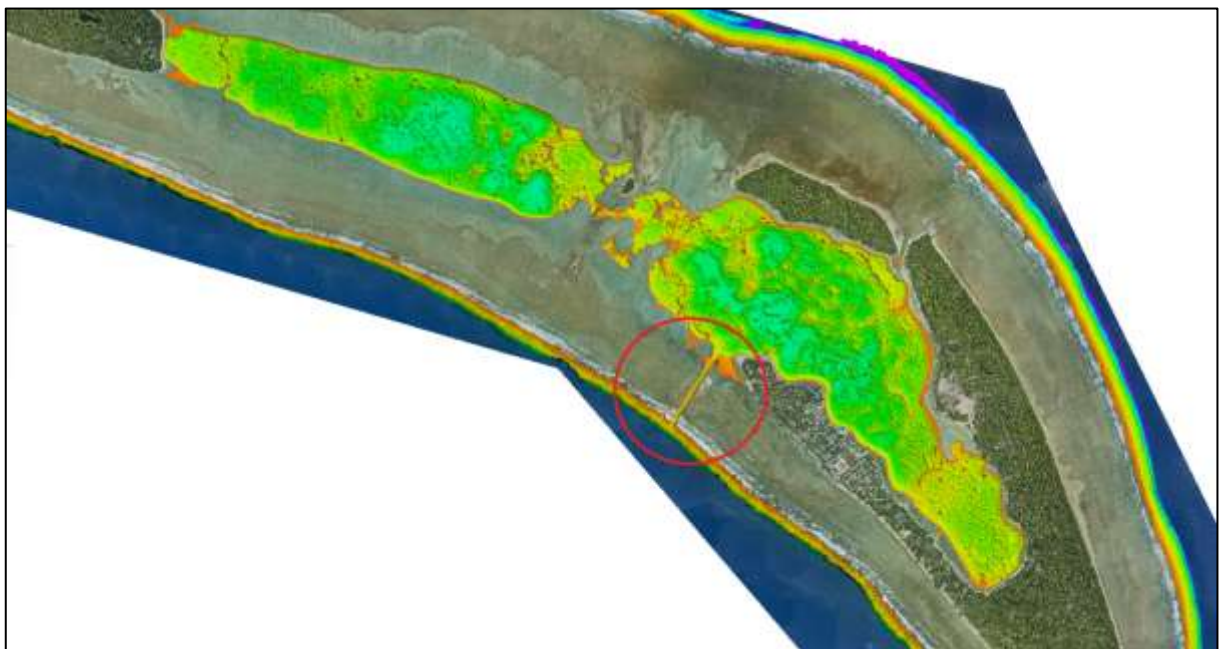


Figure 4.1: Nanumea Atoll Lagoons; Channel (Red) and Numerous Coral Pinnacles Within

Although the interior of the atoll has some distinct regions of flat reef platform it predominantly extends from the shoreline through the intertidal zone, to a maximum depth of approximately 20m where the LADS HD system has achieved near full coverage.

Strewn throughout both lagoons are numerous coral pinnacles. These pinnacles vary in diameter, from 5m to 50m, and are typically capped by coral reef. These pinnacles can pose a significant risk to safety of navigation as they often extend from the seafloor to 5m below the surface of the water.

Nanumea achieved excellent results from both ALB systems, with the LADS HD system achieving depths to +45m. The water clarity and conditions were very favourable. The shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally.



Some minor gaps do occur around the reef perimeter in areas of localised white water. There is also an area of localised turbidity within the South-eastern extent of the lagoon leaving a gap in seabed coverage. See Figure 4.2 below for final coverage.

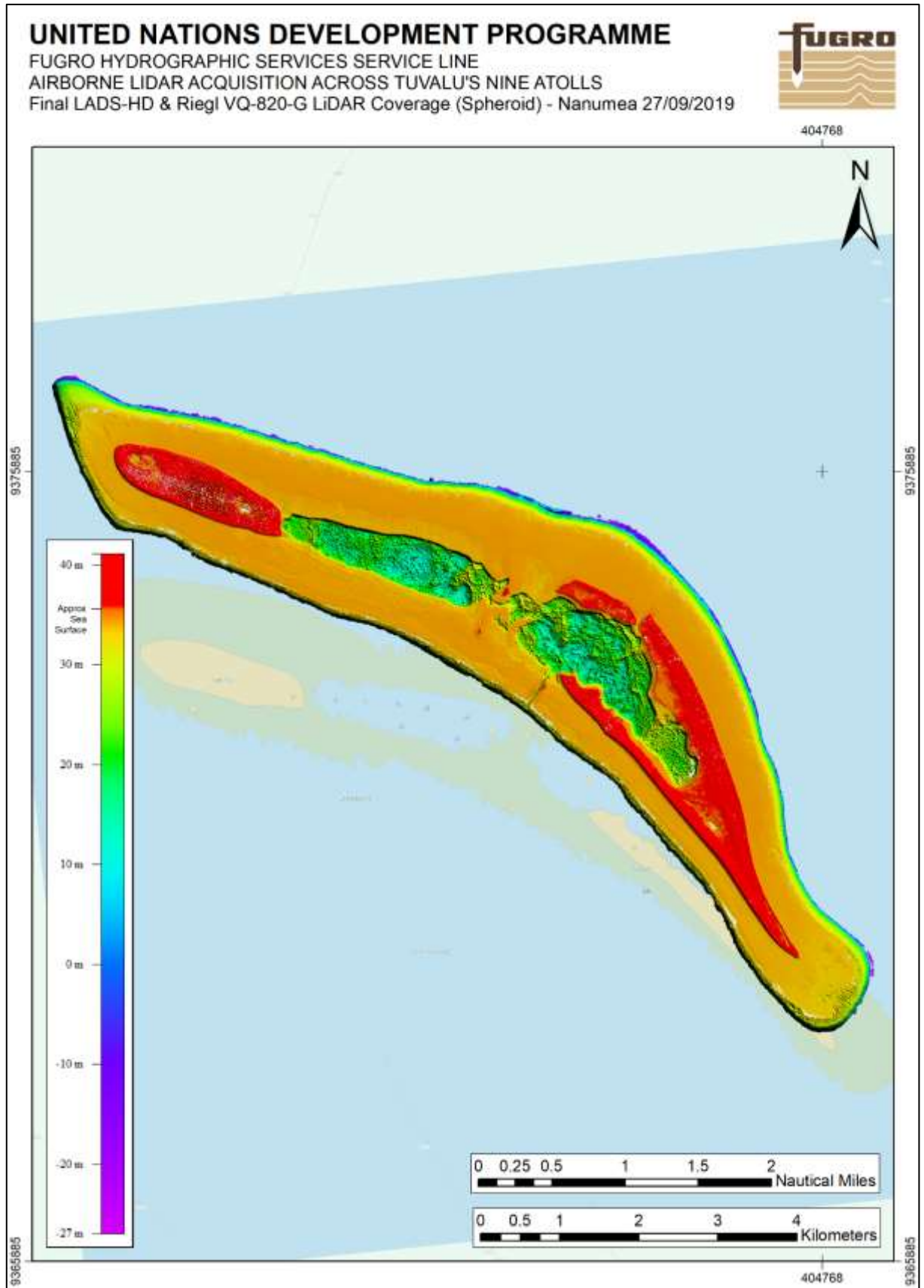


Figure 4.2: Final Survey Coverage - Nanumea

4.2.2 Orthorectified Imagery

The Nanumea Atoll encountered white water and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.

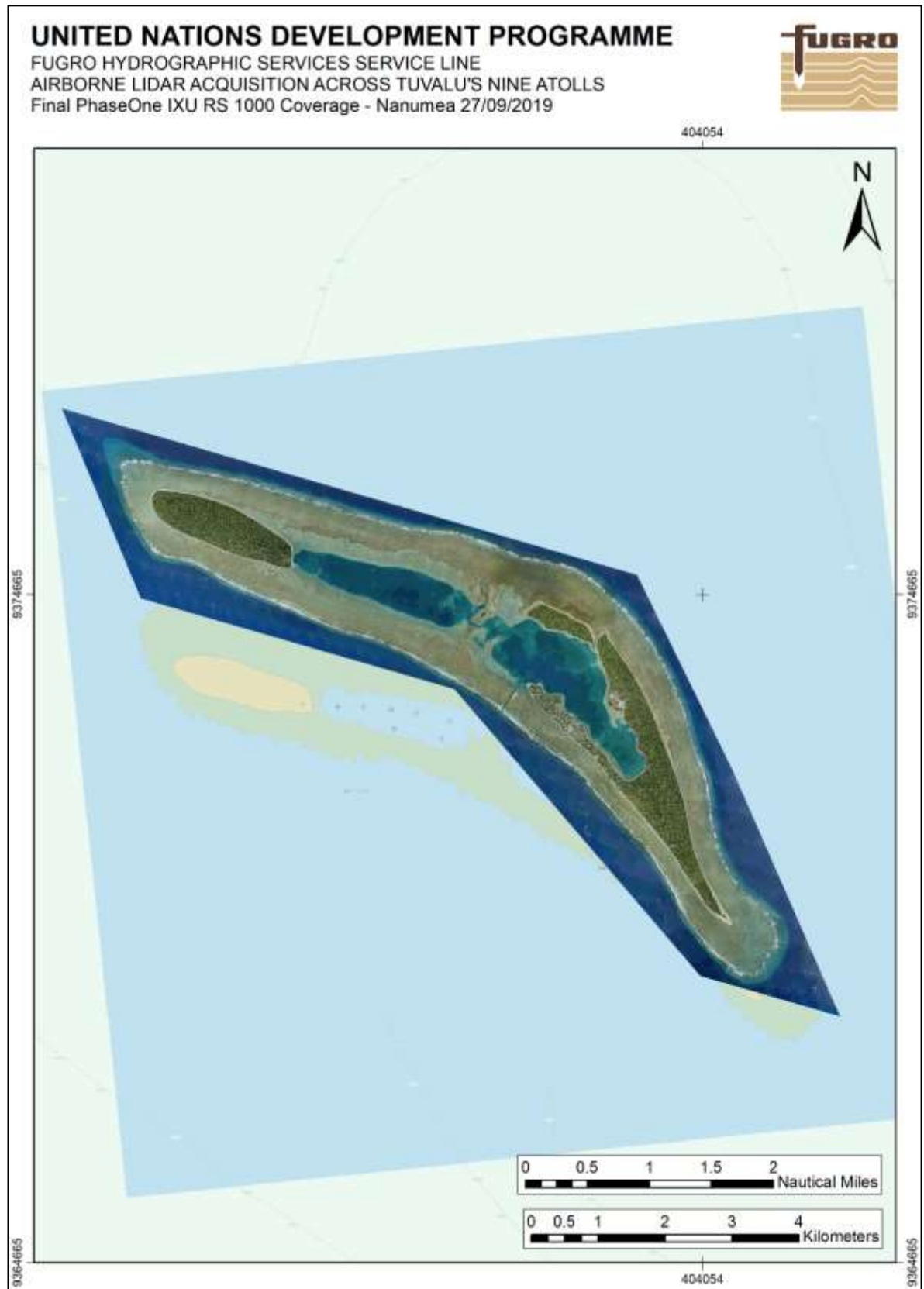


Figure 4.3: Final RGB Orthorectified Mosaic - Nanumea

4.3 Nanumaga

4.3.1 ALB Coverage

Nanumaga is the second most North-Western Atoll of Tuvalu and made up of the remains of calcareous reef-forming organisms encircling a small shallow lagoon.

Extending from the high waterline, generally out to 50m or more, a shallow sandy / reef platform encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over a thousand meters. Approximately 1.8Nm to the north of the main atoll is also a shallow reef of approximately 12m in depth which could pose a navigational hazard for larger surface vessels.

Nanumaga has one narrow shallow channel for surface vessels on the western side of the atoll to access the main township, as shown in Figure 4.4.



Figure 4.4: Nanumaga Atoll; Shallow Channel (Red)

Nanumaga achieved excellent results from both ALB systems, with the LADS HD system achieving depths to +45m due to favourable water clarity and conditions. The shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally.

Some minor gaps do occur around the reef perimeter in areas of localised white water including the exposed southern reef tip of Nanumaga. Gaps also exist within the perimeter of the lagoon itself as a result of dense vegetation. See Figure 4.5 below for final coverage.

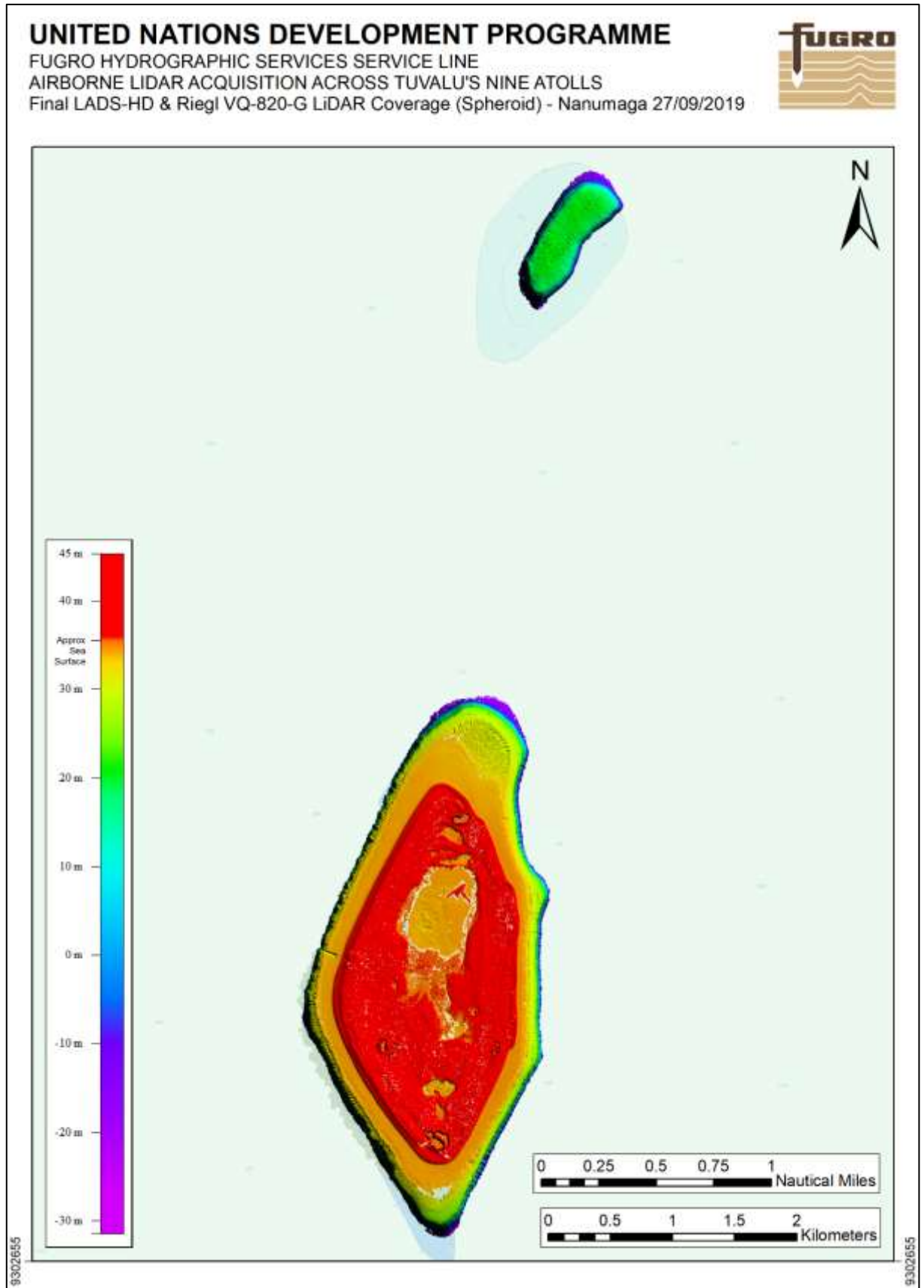


Figure 4.5: Final Survey Coverage - Nanumaga

4.3.2 Orthorectified Imagery

The Nanumaga Atoll encountered white water and cloud shadows. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop

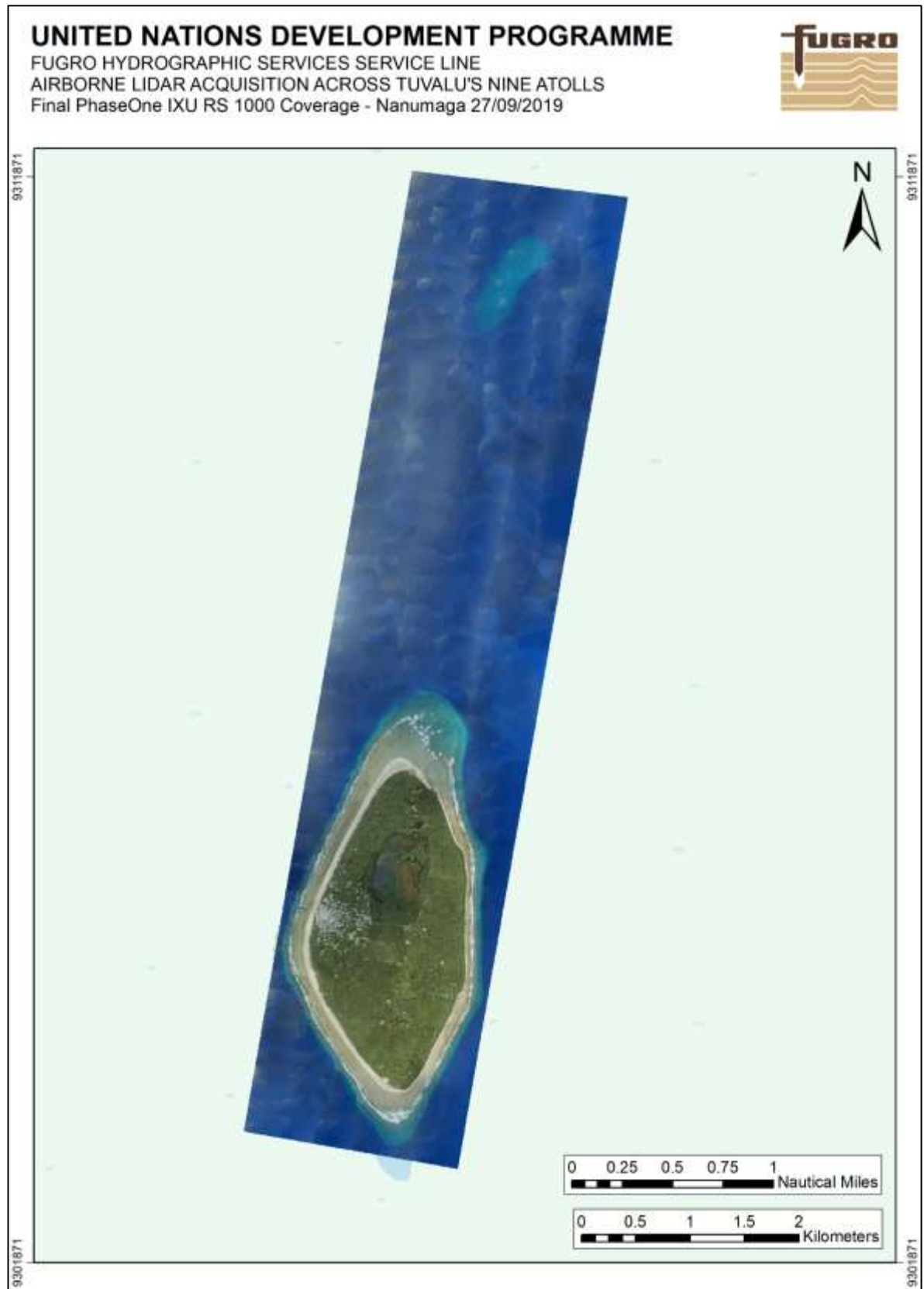


Figure 4.6: Final RGB Orthorectified Mosaic - Nanumaga

4.4 Niutao

4.4.1 ALB Coverage

Niutao Atoll is located East of the Nanumea and Nanumaga Atolls in the North of Tuvalu and made up of the remains of calcareous reef-forming organisms encircling a small shallow lagoon.

Extending from the high waterline, generally out to 100m, a shallow sandy / reef platform encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over a thousand meters.

Niutao has one small shallow channel for surface vessels on the north-west approach to the atoll to access the main township, as shown in Figure 4.11.



Figure 4.7: Niutao Atoll; Shallow Channel (Red)

Niutao achieved excellent results from both ALB systems, with the LADS HD system achieving depths to +45m. The water clarity and conditions were very favourable. The shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally.

Some minor gaps do occur around the reef perimeter in areas of localised white water including the North-eastern and South Eastern atoll tips. Gaps also exist within the perimeter of the lagoon itself as a result of dense vegetation. See Figure 4.8 below for final coverage.

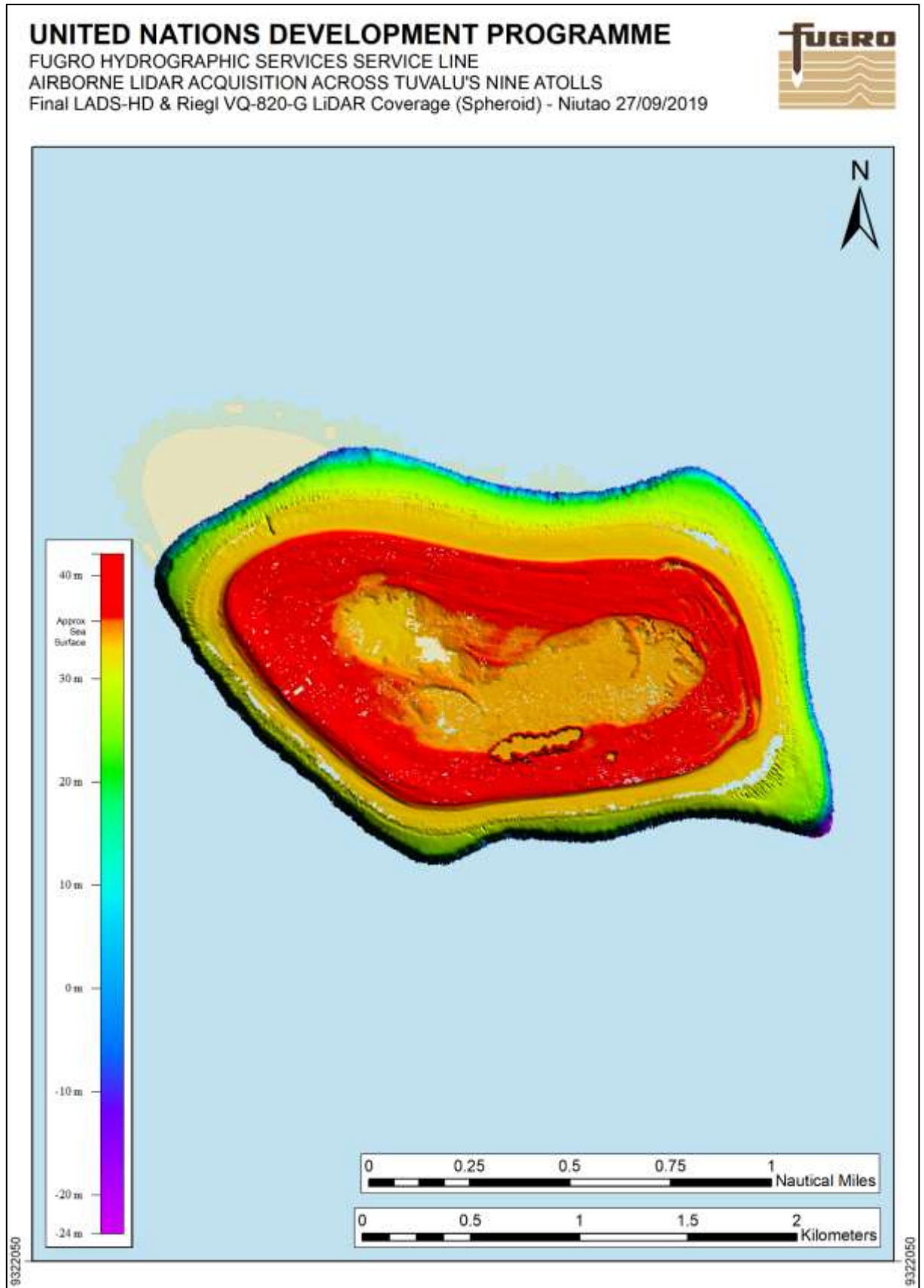


Figure 4.8: Final Survey Coverage - Niutao

4.4.2 Orthorectified Imagery

The Niutao Atoll encountered cloud shadows and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.



Figure 4.9: Final RGB Orthorectified Mosaic - Niutao

4.5 Nui

4.5.1 ALB Coverage

Nui Atoll is located central-West of Tuvalu and made up of the remains of calcareous reef-forming organisms encircling a central lagoon.

Extending from the high waterline, generally out to 300m or more, a shallow reef platform comprised of rocky outcrops and hard coral flats encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over a thousand meters.

Nui has one distinct lagoon at the centre of the atoll. There is single shallow channel for surface vessels to access the main township in the south-west of the atoll, however unlike most other atolls in Tuvalu, there is no exterior access to the central lagoon itself, as shown in Figure 4.11.



Figure 4.10: Nui Atoll; Channel (Red) and Numerous Coral Pinnacles Within the Lagoon

Although the interior of the atoll has some distinct regions of flat reef platform it predominantly extends from the shoreline through the intertidal zone, to a maximum depth of approximately 15m where the LADS HD system has achieved full coverage.

Strewn throughout the lagoon are numerous coral pinnacles especially on the northern end. These pinnacles vary in diameter, from 5m to 100m, and are typically capped by coral reef. While vessel access is minimal these pinnacles could pose a significant risk to safety of navigation as they often extend from the seafloor to 5m below the surface of the water. See Figure 4.10.

Nui achieved excellent results from both ALB systems, with the LADS HD system achieving depths to +45m. The water clarity and conditions were very favourable. The shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally.

Some minor gaps do occur around the reef perimeter in areas of localised white water. See Figure 4.11 below for final coverage.

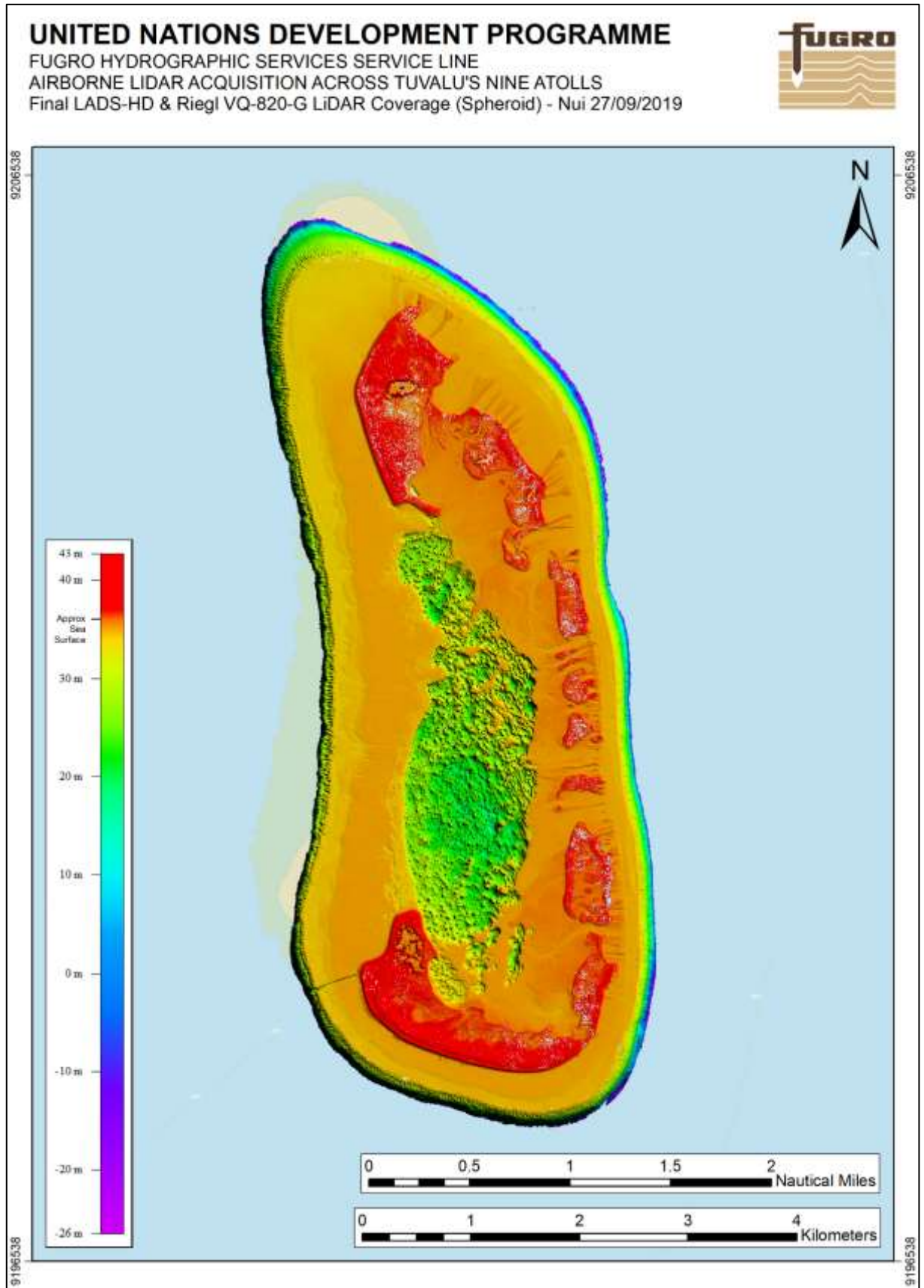


Figure 4.11: Final Survey Coverage – Nui

4.5.2 Orthorectified Imagery

The Nui Atoll encountered white water, cloud shadows and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.



Figure 4.12: Final RGB Orthorectified Mosaic - Nui

4.6 Vaitupu

4.6.1 ALB Coverage

Vaitupu Atoll is located central-East of Tuvalu and made up of the remains of calcareous reef-forming organisms encircling a central small lagoon.

Extending from the high waterline, generally out to 100m or more, a shallow sandy / reef platform encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over a thousand meters.

Vaitupu has one distinct lagoon at the centre of the atoll. There is single shallow channel for surface vessels to access the main township in the south-west of the atoll, however unlike most other atolls in Tuvalu, there is no exterior access to the central lagoon itself, as shown in Figure 4.15.

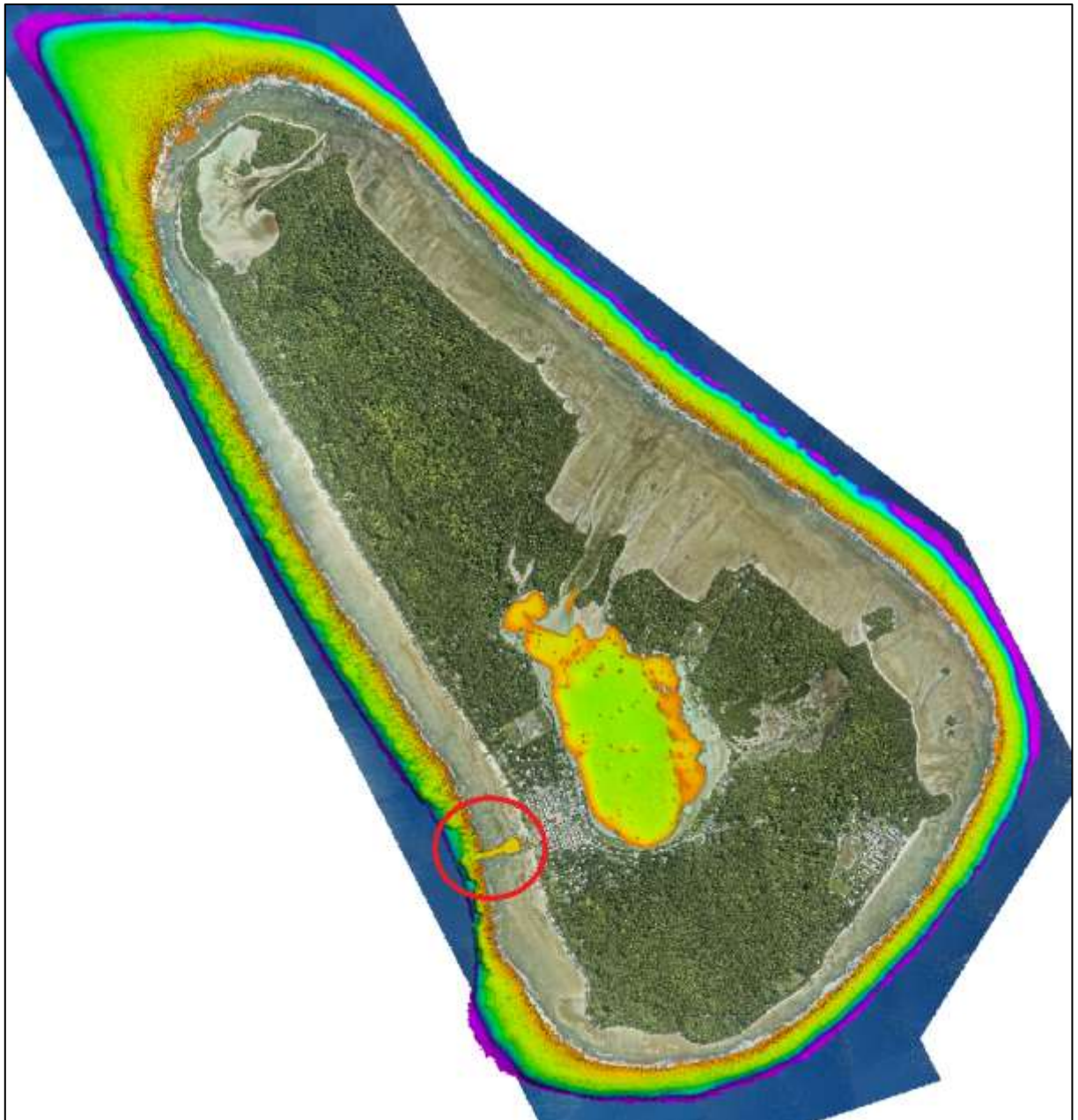


Figure 4.13: Vaitupu Atoll; Channel (Red) and Numerous Coral Pinnacles Within the Lagoon

Although the interior of the atoll has some distinct regions of flat reef platform it predominantly extends from the shoreline through the intertidal zone, to a maximum depth of approximately 10m where the LADS HD system has achieved full coverage.

Within the lagoon are numerous coral pinnacles. These pinnacles vary in diameter, from 5m to 50m, and are typically capped by coral reef. While vessel access is minimal these pinnacles could pose a significant risk to safety of navigation as they often extend from the seafloor to 5m below the surface of the water. See Figure 4.13.

Vaitupu achieved excellent results from both ALB systems, with the LADS HD system achieving depths to +50m. The water clarity and conditions were very favourable. The shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally.

Some minor gaps do occur around the reef perimeter in areas of localised white water especially on the exposed southern and eastern side of the atoll. Within the lagoon itself are large areas of turbidity resulting in gaps in coverage especially on the northern end. See Figure 4.14 below for final coverage.

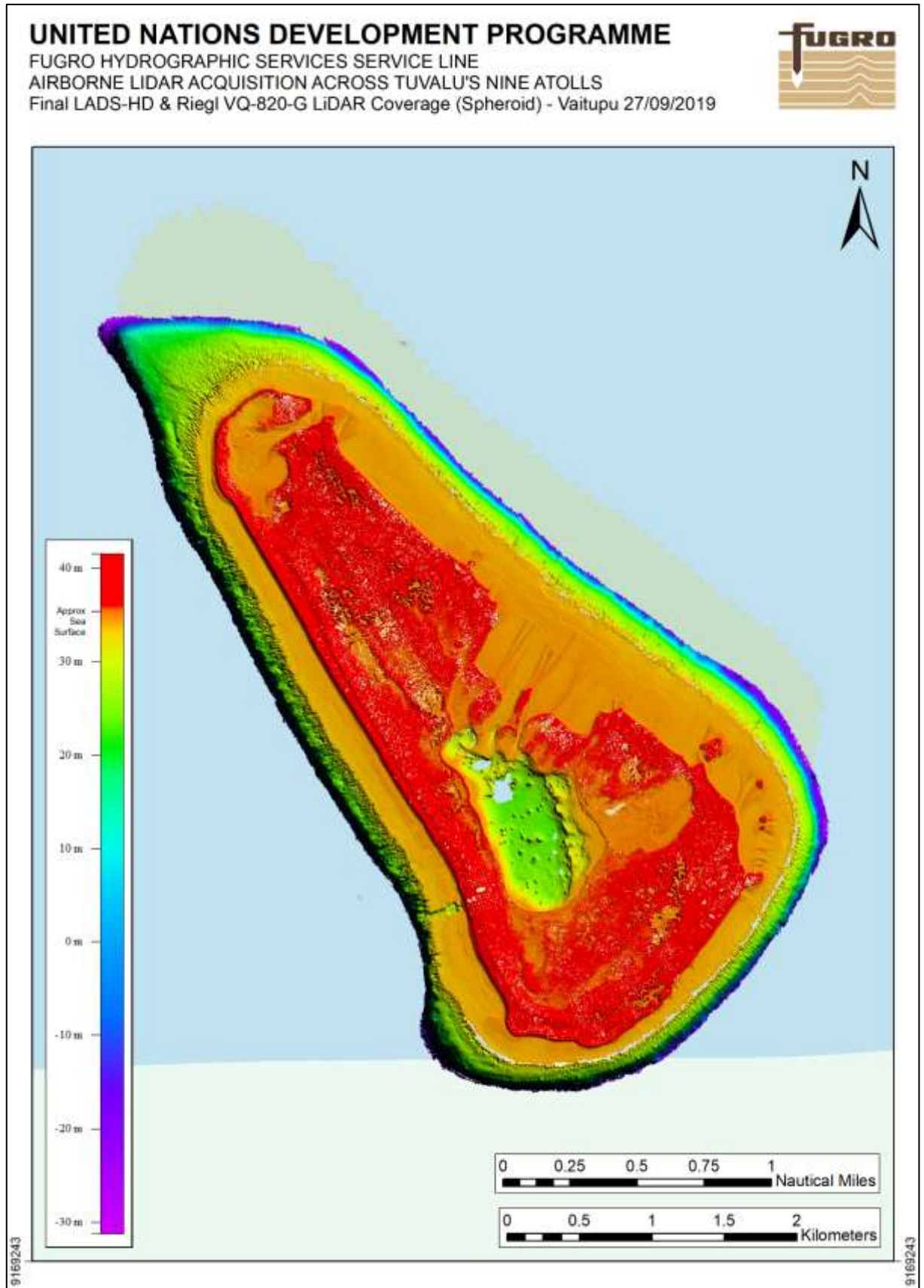


Figure 4.14: Final Survey Coverage – Vaitupu

4.6.2 Orthorectified Imagery

The Vaitupu Atoll encountered cloud shadows and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.



Figure 4.15: Final RGB Orthorectified Mosaic – Vaitupu

4.7 Nukufetau

4.7.1 ALB Coverage

Nukufetau is located just North-West of Funafuti and the second largest atoll in Tuvalu. The topography of Nukufetau Atoll is made up of the remains of calcareous reef-forming organisms encircling a central lagoon.

Extending from the high waterline, generally out to 100m, a shallow reef platform comprised of rocky outcrops and hard coral flats encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over two thousand meters in places.

The main access points for surface vessels into the Nukufetau lagoon is through a single deep channel on the western side of Nukufetau as shown in Figure 4.20.

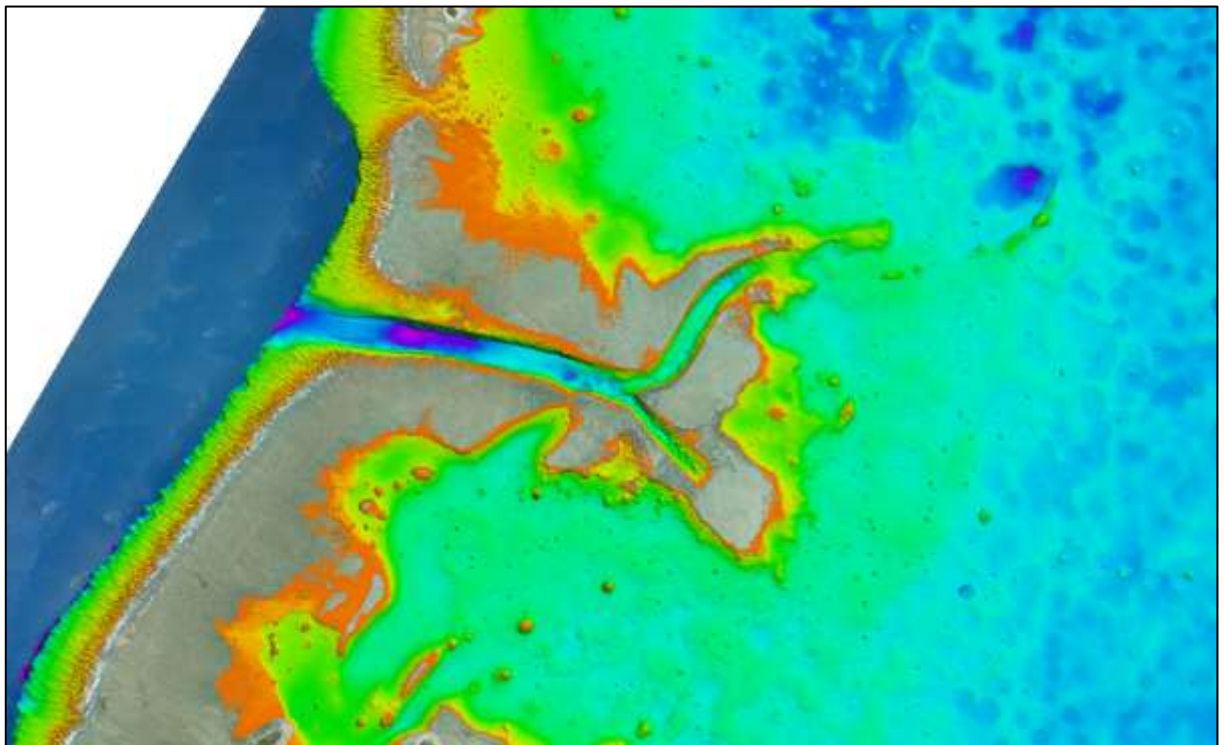


Figure 4.16: Nukufetau Atoll Western Access Channel with Shallow Reef and Pinnacles Inside the Lagoon

Strewn throughout the lagoon are also numerous coral pinnacles. These pinnacles vary in diameter, from 50m to greater than 200m, and are typically capped by coral reef. These pinnacles can pose a significant risk to safety of navigation as they often extend from the seafloor to 5m below the surface of the water.

Although the interior of the atoll still has distinct regions of flat reef platform extending from the shoreline through the intertidal zone, the near shore bathymetry within the lagoon can better be characterised as a moderately sloping sandy seabed covered with numerous coral heads and rock outcrops. Beyond the

near shore bathymetry within the lagoon, most of the seabed exhibits a rough and undulating texture documented in previous surveys to be Halimeda mounds, see Figure 4.21.

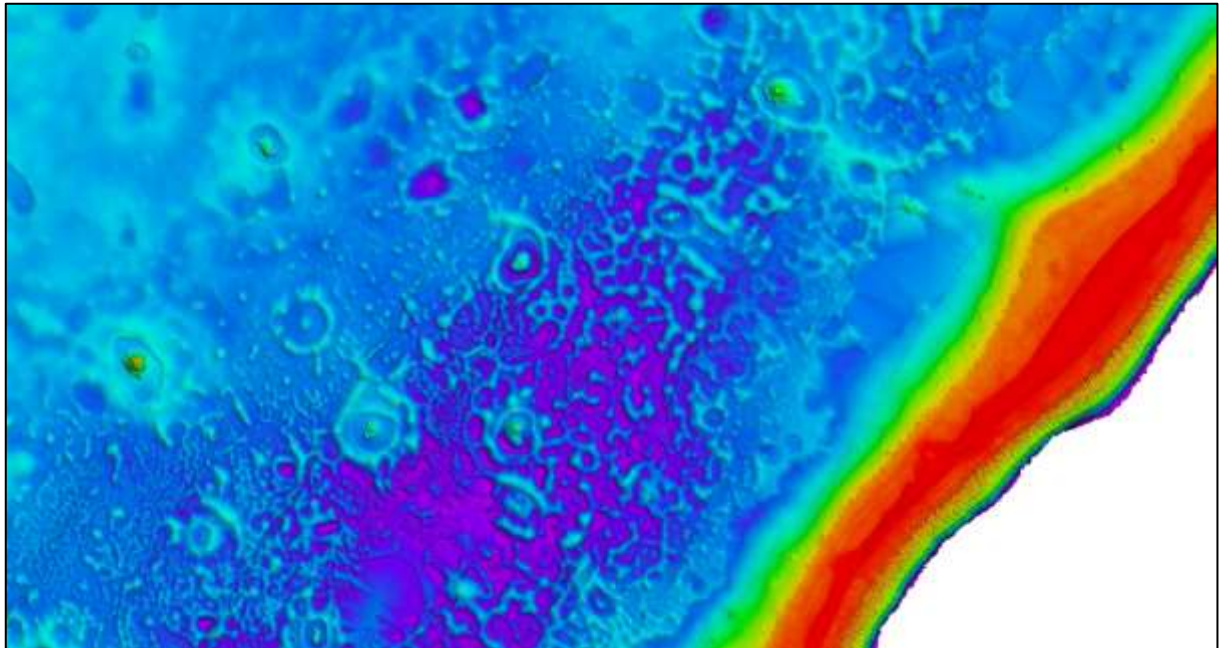


Figure 4.17: Example of Halimeda Mounds Along Eastern Side of Nukufetau Lagoon

In deeper regions of the lagoon, particularly along the eastern side, are areas where the seabed has not been surveyed. This was due to consistent turbidity and in cases beyond the extinction depth of the LADS HD system, which nominally was beyond 40m depth. In the nearshore regions the shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally. Overall coverage though throughout Nukufetau was excellent.

Some other minor gaps do occur around the reef perimeter in areas of localised white water. See Figure 4.18 below.

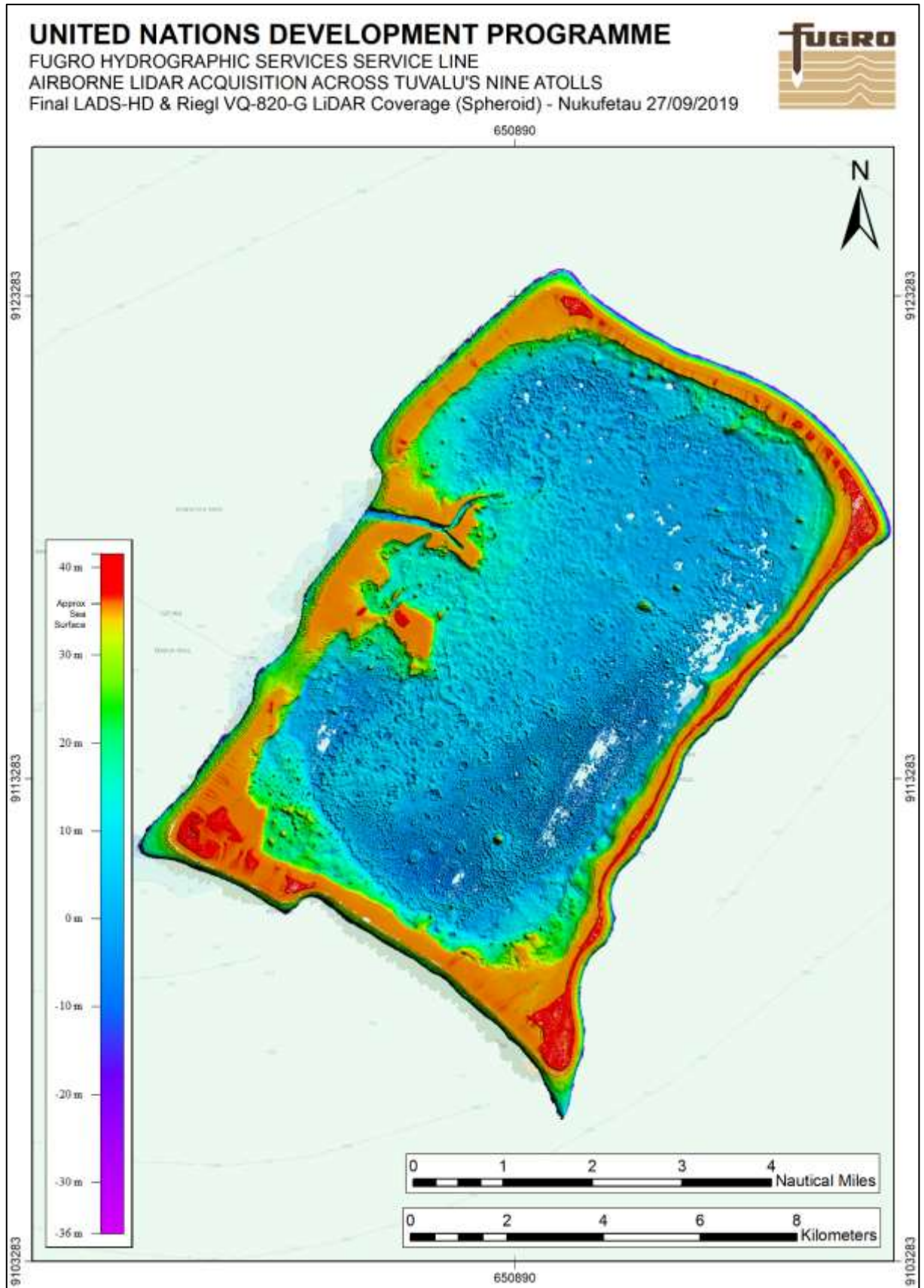


Figure 4.18: Final Survey Coverage – Nukufetau

4.7.2 Orthorectified Imagery

The Nukufetau Atoll encountered cloud shadows, white water and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.

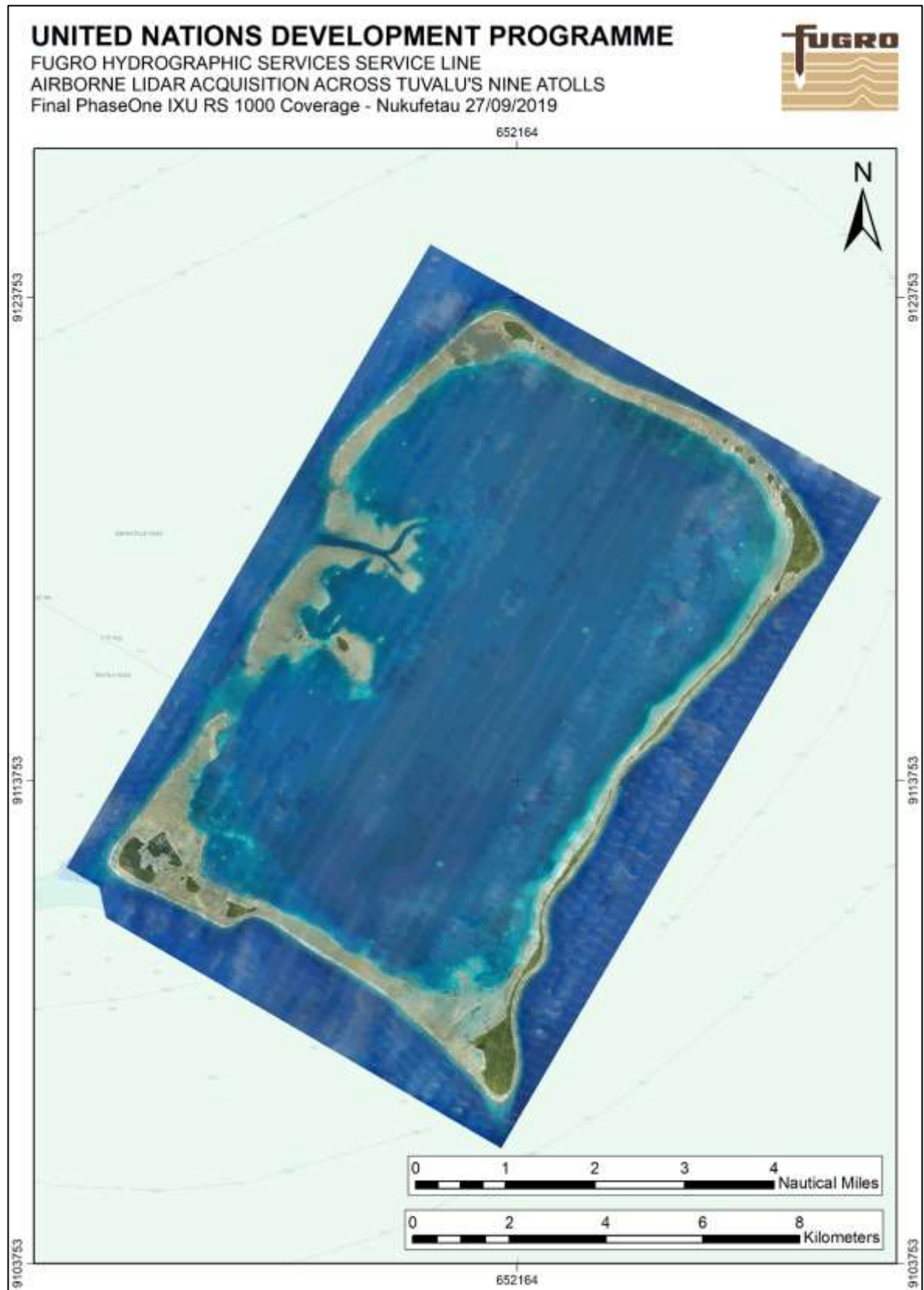


Figure 4.19: Final RGB Orthorectified Mosaic - Nukufetau

4.8 Funafuti

4.8.1 ALB Coverage

Funafuti is the largest and most prominent Atoll of Tuvalu and typical of most atoll environments of the Pacific region. The topography of Funafuti Atoll is made up of the remains of calcareous reef-forming organisms encircling a central lagoon.

Extending from the high waterline, generally out to 120m, a shallow reef platform comprised of rocky outcrops and hard coral flats encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over two thousand meters in places.

The main access points for surface vessels into the Funafuti lagoon is through two deep channels along the western side of Funafuti near the islands of Fuagea and Tepuka as shown in Figure 4.20.

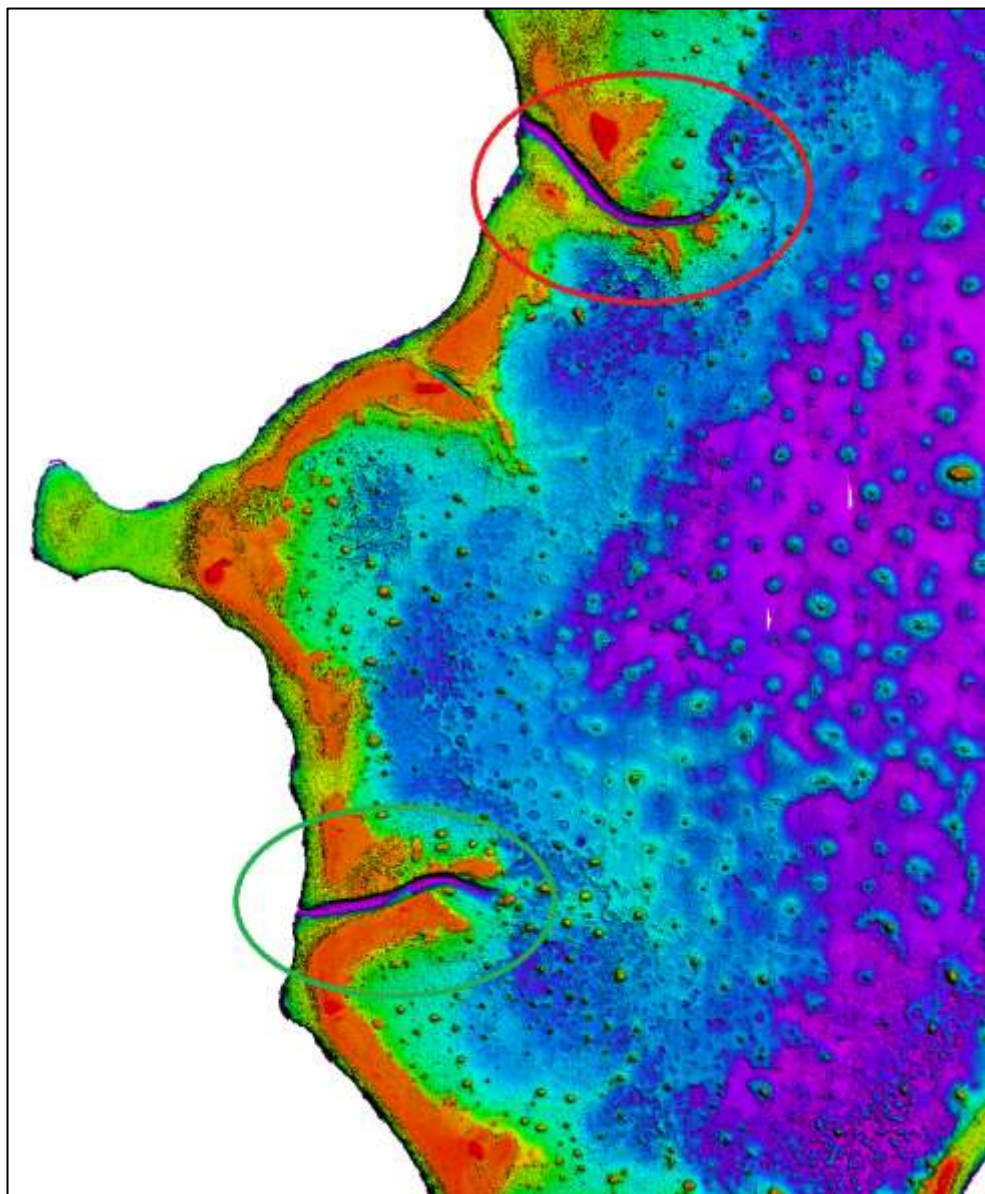


Figure 4.20: Funafuti Atoll Western Access Channels: Tepuka Channel (Red) and Fuagea Channel (Green) in Region of Numerous Coral Pinnacles

Strewn throughout the lagoon, with a greater concentration in the western half, are coral pinnacles. These pinnacles vary in diameter, from 50m to greater than 200m, and are typically capped by coral reef. These pinnacles can pose a significant risk to safety of navigation as they often extend from the seafloor to 5m below the surface of the water. See Figure 4.20.

Although the interior of the atoll still has distinct regions of flat reef platform extending from the shoreline through the intertidal zone, the near shore bathymetry within the lagoon can better be characterised as a moderately sloping sandy seabed covered with numerous coral heads and rock outcrops. Beyond the near shore bathymetry within the lagoon, the vast majority of the seabed exhibits a rough and undulating texture documented in previous surveys to be Halimeda mounds, see Figure 4.21.

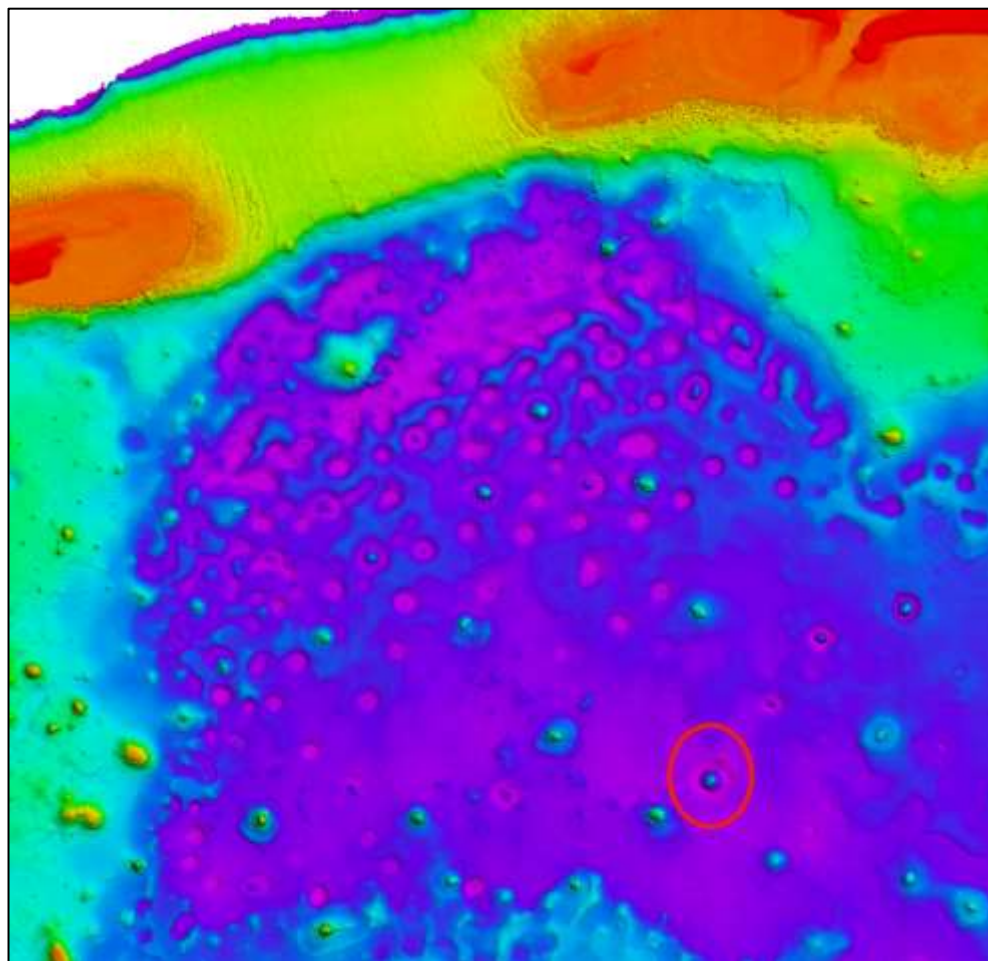


Figure 4.21: Example of Halimeda Mounds (Red) to the NW of Funafuti Lagoon

In deeper regions of the lagoon there are large areas where the seabed has not been surveyed as it was quite turbid and beyond the extinction depth of the LADS HD system, generally beyond 40m depth. In the nearshore regions the shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally. Overall coverage though throughout Funafuti was excellent.

Some minor gaps do occur around the reef perimeter in areas of localised white water. The water clarity and conditions were generally favourable however, like Nukufetau, in the deeper depressions within the Funafuti lagoon itself generally remained quite turbid. These areas were reflowed a number of times to improve coverage with limited success. See Figure 4.22 below for final coverage.

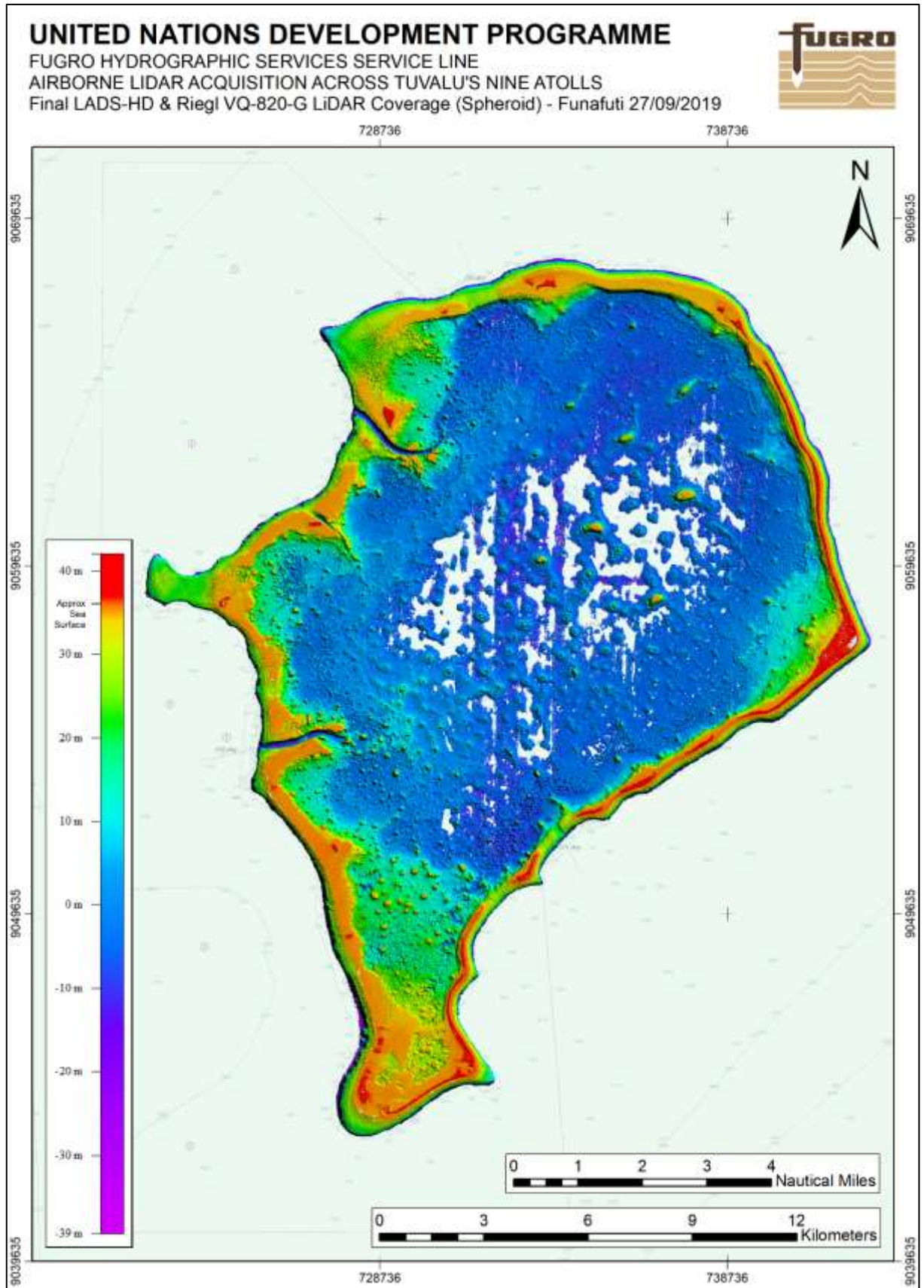


Figure 4.22: Final Survey Coverage – Funafuti

4.8.2 Orthorectified Imagery

The Funafuti Atoll encountered cloud shadows, white water and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.

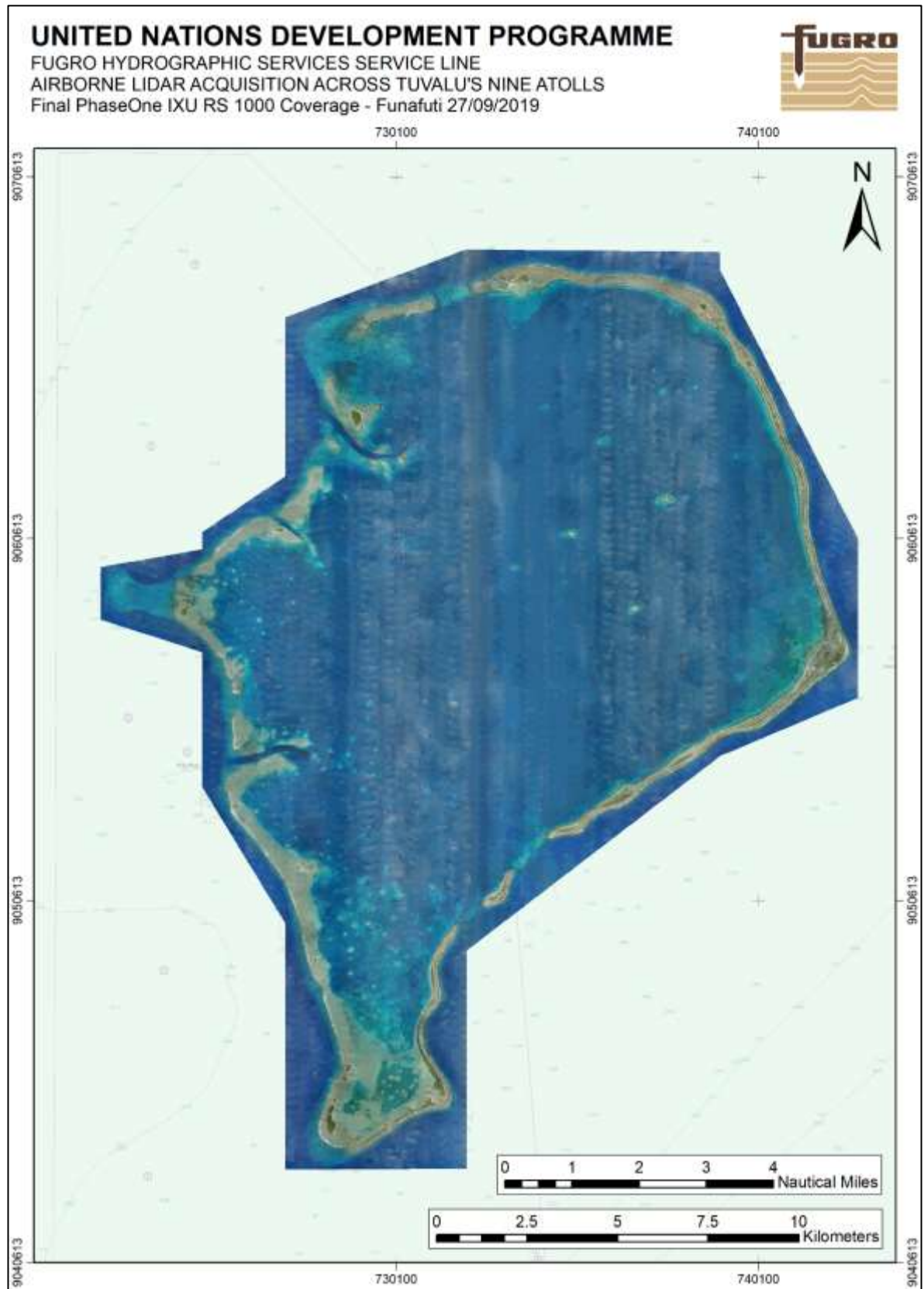


Figure 4.23: Final RGB Orthorectified Mosaic – Funafuti

4.9 Nukulaelae

4.9.1 ALB Coverage

Nukulaelae is located South-East of Funafuti and made up of the remains of calcareous reef-forming organisms encircling a central lagoon.

Extending from the high waterline, generally out to 100m or more, a shallow reef platform comprised of rocky outcrops and hard coral flats encompasses the atoll. This gently sloping platform reaches an average depth of approximately 10m at its furthest point from the high waterline before steeply dropping off to well over a thousand meters.

Nukulaelae has one distinct lagoon at the centre of the atoll. There is a single shallow channel for surface vessels to access the main township to the north-west of the atoll, however there is no exterior access to the central lagoon itself, as shown in Figure 4.24 and Figure 4.25.

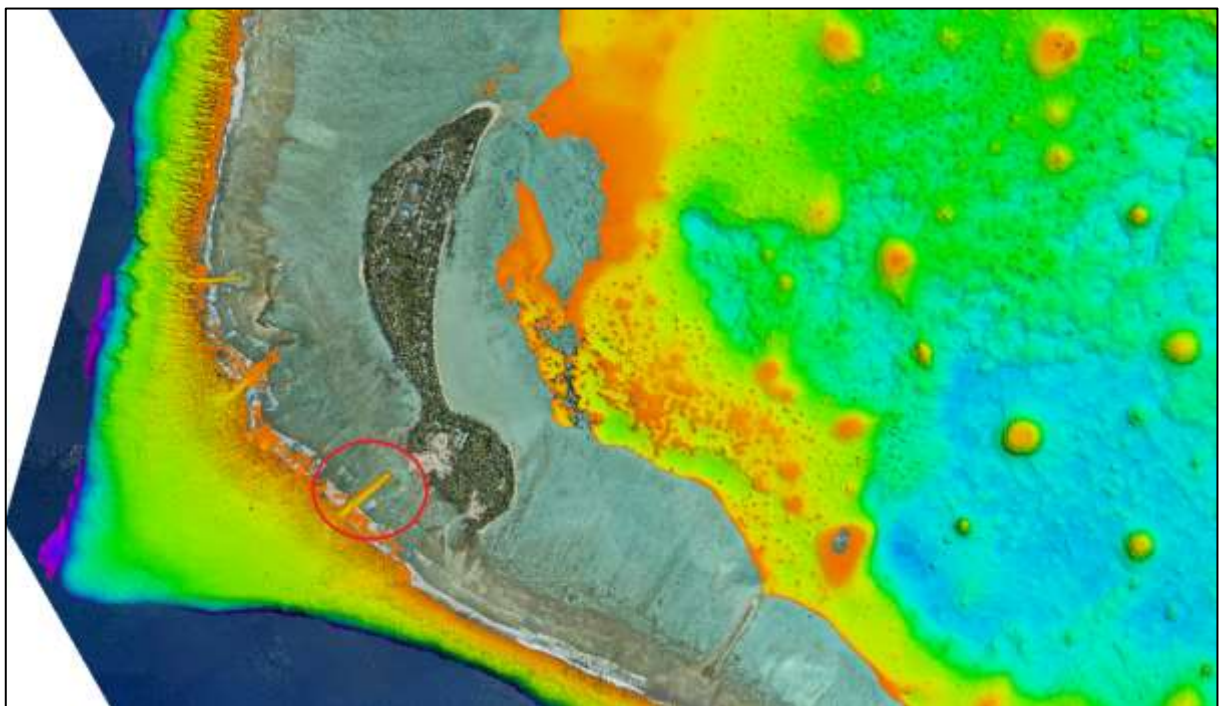


Figure 4.24: Nukulaelae Atoll; Channel (Red) and Numerous Coral Pinnacles Within the Lagoon

Although the interior of the atoll still has distinct regions of flat reef platform, extending from the shoreline through the intertidal zone, the near shore bathymetry within the lagoon can better be characterised as a moderately sloping sandy seabed covered with numerous coral heads and rock outcrops. Beyond the near shore bathymetry within the lagoon, the vast majority of the seabed exhibits a rough and undulating texture, see Figure 4.24.

Strewn throughout the lagoon are numerous coral pinnacles. These pinnacles vary in diameter, from 5m to 160m, and are typically capped by coral reef. While vessel access is minimal these pinnacles could pose a significant risk to safety of navigation as they often extend from the seafloor to 5m below the surface of the water as shown in Figure 4.24 also.

In deeper regions of the lagoon, to the north, are areas where the seabed has not been surveyed. This was due to consistent turbidity and in cases just beyond the extinction depth of the LADS HD system, which nominally was 30m depth. This area was re flown with some improved coverage. In the nearshore regions the shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally. Overall coverage though throughout Nukulaelae was excellent.

Some other minor gaps do occur around the reef perimeter in areas of localised white water especially along the southern and western extents. See Figure 4.25 below for final coverage.

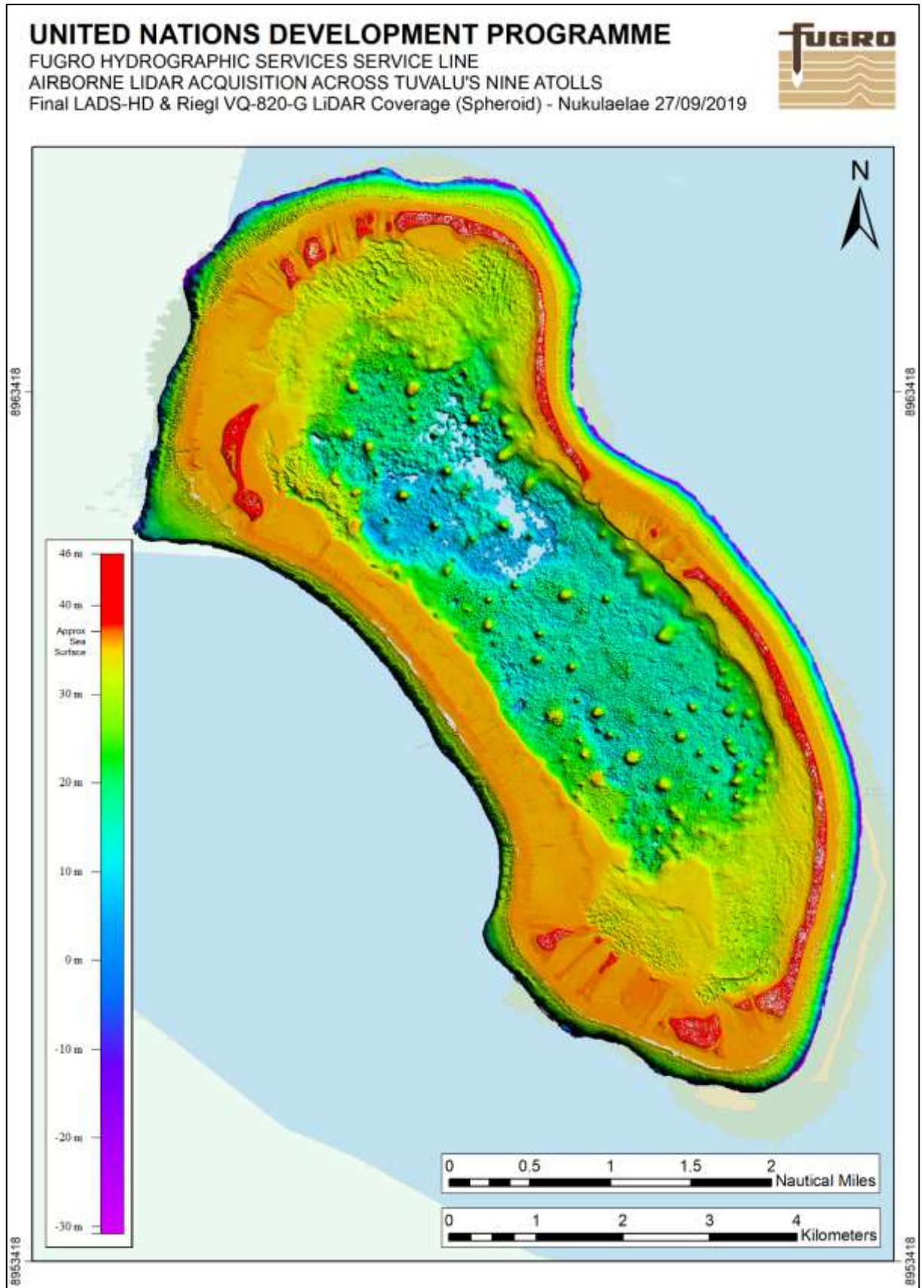


Figure 4.25: Final Survey Coverage – Nukulaelae

4.9.2 Orthorectified Imagery

The Nukulaelae Atoll encountered white water and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.



Figure 4.26: Final RGB Orthorectified Mosaic – Nukulaelae

4.10 Niulakita

4.10.1 ALB Coverage

Niulakita is the most Southern Atoll of Tuvalu and made up of the remains of calcareous reef-forming organisms and is part of a larger seamount system as explored with the investigation lines shown in section 4.11.

Extending from the high waterline, generally out to 100m, a shallow sandy / reef platform encompasses the atoll. This gently sloping platform reaches an average depth of approximately 30m at its furthest point from the high waterline before steeply dropping off to well over a thousand meters to the west. To the east it extends for another 5 Nm as part of a larger sea mount system before steeply dropping off, as shown in the investigation line coverage of Figure 4.29.

Niulakita overall achieved excellent results from both ALB systems, with the LADS HD system achieving depths to +45m. Unfortunately, the contracted survey area design restricted the fulfilment of reaching the LADS HD extinction depths to the north, east and south especially. The shallow water RIEGL VQ-820-G system reached an extinction depth to 10m generally.

Some minor gaps do occur around the reef perimeter in areas of localised white water especially along the southern extents. See Figure 4.27 below.

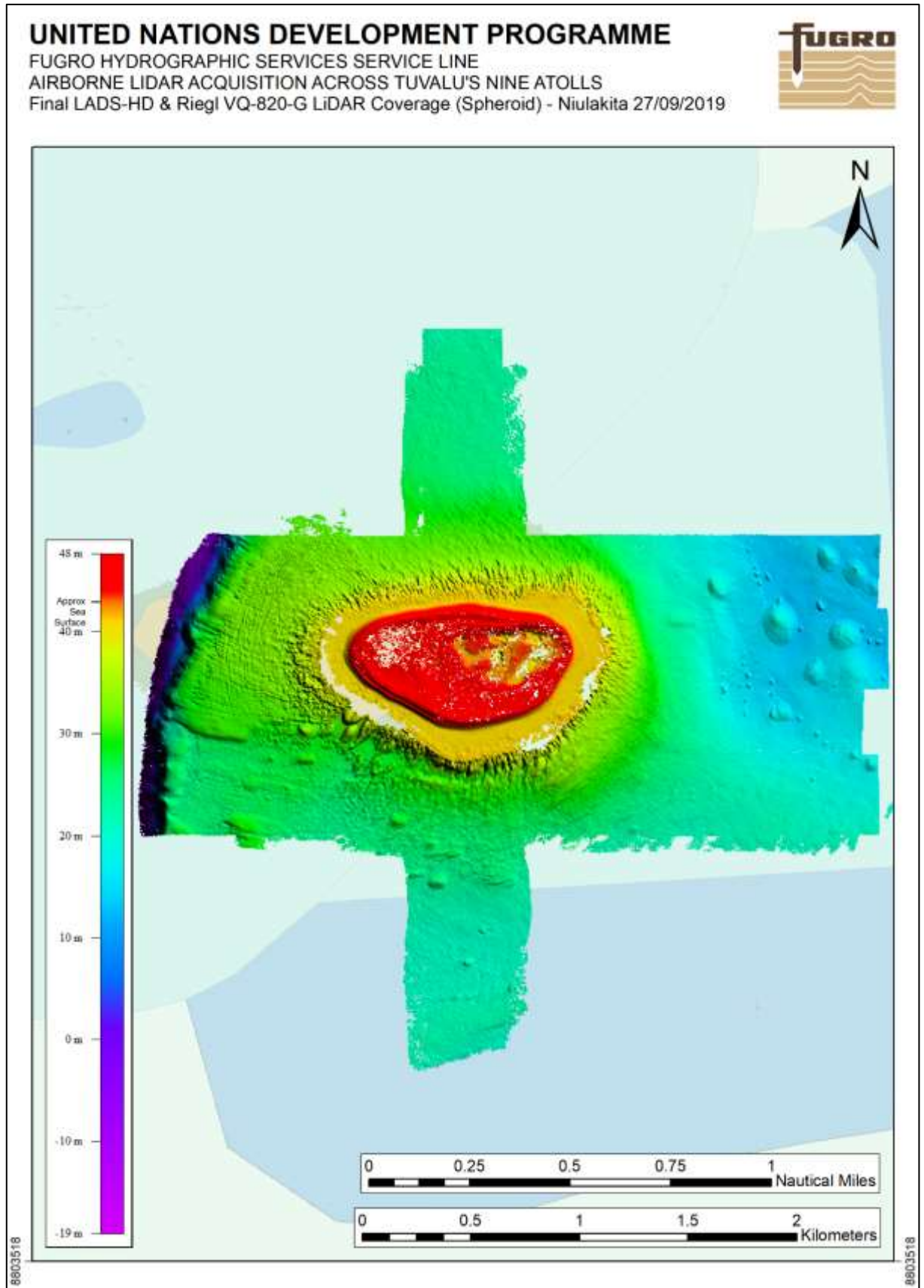


Figure 4.27: Final Survey Coverage – Niulakita

4.10.2 Orthorectified Imagery

The Niulakita Atoll encountered white water and sun glint. Best care was taken to correct these issues along with major colour differences using seamline editing and Photoshop.

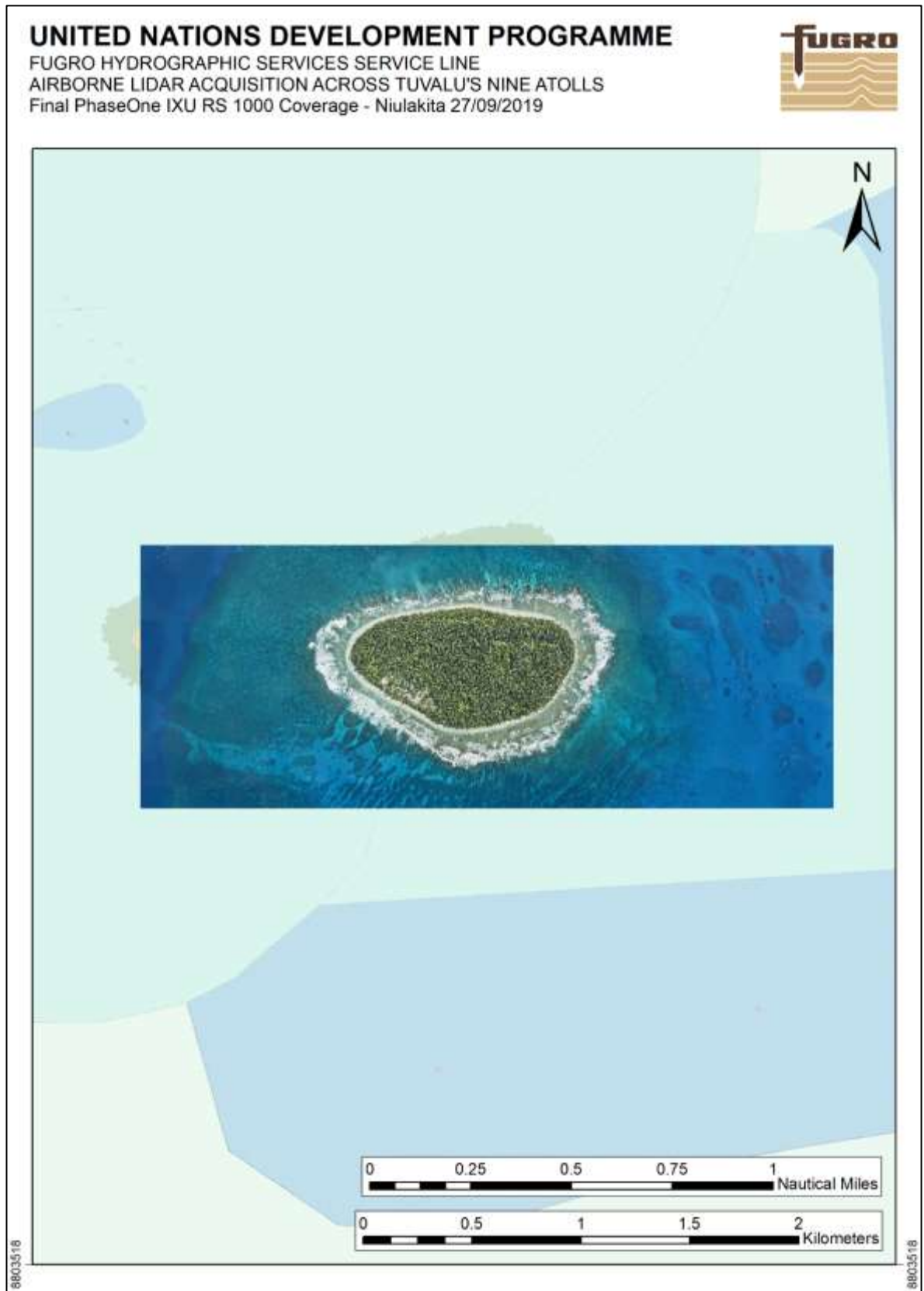


Figure 4.28: Final RGB Orthorectified Mosaic – Niulakita

4.11 Investigation Area near Niulakita

In discussion with the client during the data collection phase it was agreed to conduct seven extra investigation lines around Niulakita to detect the extents of the seamounts. The seamounts extended approximately 50km north and 40km east of Niulakita and were generally in depths of 20m or deeper.

The water clarity and conditions were excellent in this region resulting with the LADS HD system achieving solid coverage to depths of +50m. No coverage was achieved with the shallow water RIEGL VQ-820-G system in this region as it was beyond the laser extinction depth.

In general, this region looked very promising for future work if the opportunity arose. See Figure 4.29 below for final coverage of the investigation region.

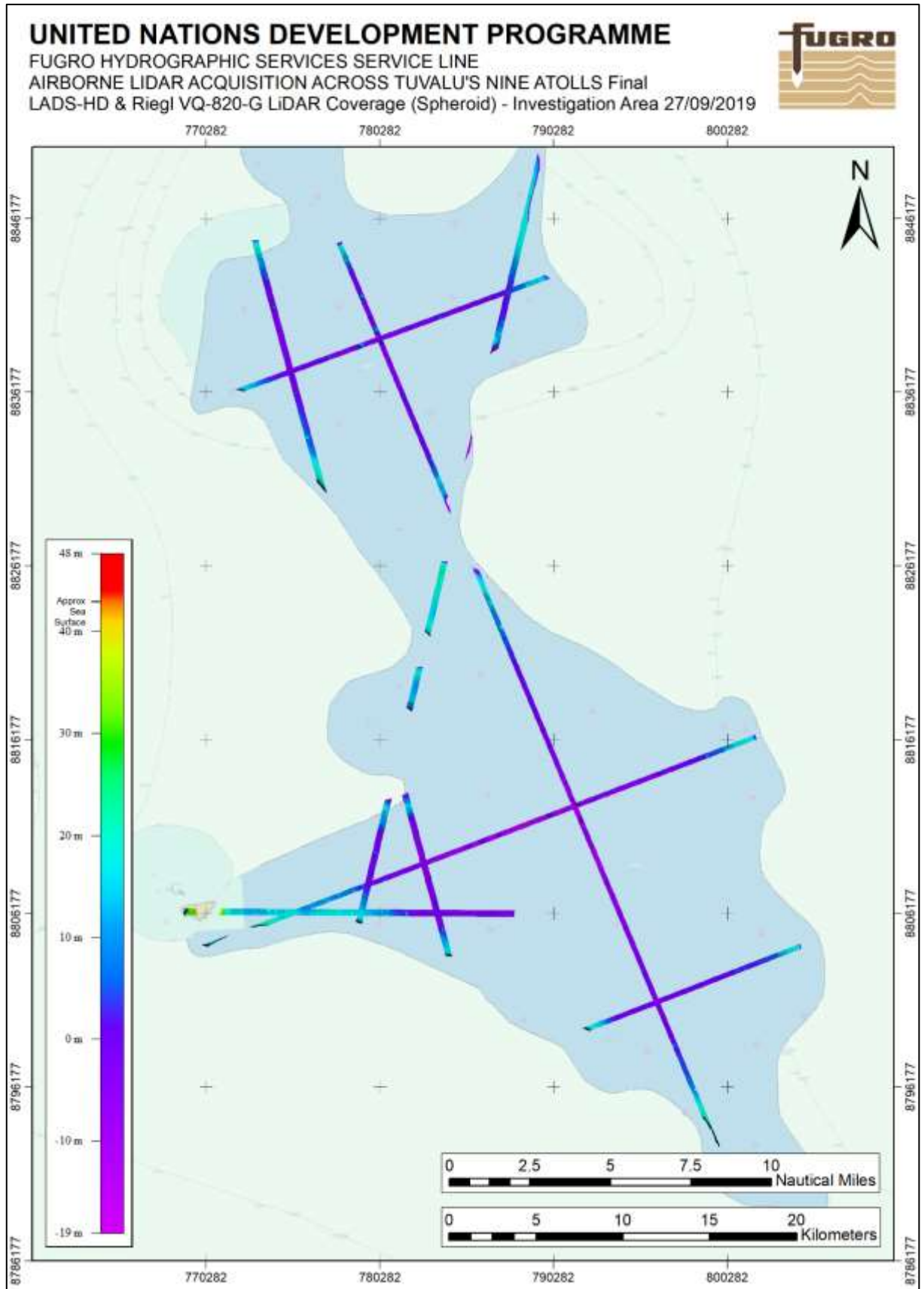


Figure 4.29: Final Survey Coverage – Investigation Area (Niulakita)

4.12 Coastline, Topography and Bathymetry

The coastline within the survey area is varied and all drying soundings over the Tuvalu Atolls have been retained as required. Topographic swath data was processed and included in the final datasets.

The data classification of ground and seabed, class 2 and 13 respectively, was done initially on the real-time sea surface as a reference. As a result of subsequent flights, collected at different tidal states, there may be some indiscriminate ground or seabed soundings within the final classified LAS datasets that have been classified either side of the true MSL zero.

All topographic data has been checked for consistency, particularly along the coastline, and any topographic data with anomalies such as datum errors have been rejected and removed from the dataset.

Any manmade breakwaters where water cannot flow underneath essentially have been classified as ground. As covered in Annex K, features such as jetties or wharfs where water can freely flow underneath have been classified as a cultural feature and may appear as a gap in the data.

All vegetation and building structures that have been detected in the survey area have not been classified and remain as "Unclassified" in accordance with the class list found in Table 3.10.

4.13 Cultural and Surveyed Features in the Data

Refer to Annex K for an overview of seabed features detected by the LADS HD and RIEGL VQ-820-G systems, including wrecks and navigational aids.

5. DELIVERABLES

A description of all data provided for the project is as follows (see annex A for more details):

Table 5.1: Project Deliverables

Deliverable	Required Deliverable	Format
Project Plan – Friday 7th December or before		
Deliverable 1	Project Execution Plan	MS Word
Mobilisation and calibration report		
Deliverable 2	This report will be provided following the completion of the sensors installation and calibration in the survey aircraft and details the results of the sensors set-to-work and calibration.	PDF
Daily Progress Reports (DPRs)		
Deliverable 3	These will be provided daily during survey operations and include a summary of survey progress in the previous 24 hours, planned survey activities for the next 24 hours and other significant operational events.	PDF
Weekly report (Data collection and data processing)		
Deliverable 4	This will include an image of the coverage achieved. Interim data in *.LAS format from the LADS sensor	GeoTiff (coverage image) LAS v1.4 for the interim LADS lidar point cloud
Report of Survey		
Deliverable 5	A Report of Survey prepared that details the survey operations and results inclusive of the following supporting documents/appendices: ■ List of Deliverables ■ Sensor Specifications ■ Geodetic Records ■ Horizontal Datum / Control ■ Vertical Datum / Control ■ Benchmark and Cross-line Comparison Results ■ Assessment of Accuracy of Soundings (Vertical and Horizontal) ■ Summary of Survey Activities, Daily Reports & Field Notes ■ Seabed Features ■ Service Warranty	PDF
Final data		

Deliverable	Required Deliverable	Format
Deliverable 6	Supplied on portable hard disk drives, Fugro will prepare the final data in the formats suitable for use by UNDP, including but not limited to: ■ All raw & processed data in native (proprietary) and LAS 1.4 formats; ■ Classified bathy data into ground & non, ground features (raw data in las format); ■ Classified point clouds provided in 1km squared tiles; ■ Digital Elevation Models – variable resolution: 1m where there is Riegl coverage (topo and shallow bathy) and 5m where there is LADS coverage (deep bathy); ■ DEM may have gaps due to water clarity or laser extinction. ■ Interpolation between points shall not be undertaken. ■ Colourised RGB lidar point cloud; ■ Orthorectified RGB imagery in ECW or GeoTiff format; ■ Other software specific formats as required. (For discussion/agreement).	Raw data: LAS v1.4 Lidar data: LAS v1.4 DEM: ASC grids Interpolated DEM: GeoTiff Imagery: ECW and tiled GeoTiff
Training		
Deliverable 7	Supplied over a one-week period the following training will be provided ■ Executive briefing – 2-hours briefing on the project. Outcome – summary of the lidar project and how the datasets will help achieve the TCAP goals. ■ Manager training – 6-hours training on the project. Outcome 1 – detailed overview of the lidar project. Outcome 2 - how the datasets will help to achieve the TCAP goals. ■ User training – 3-days training on the project. Outcome 1 – detailed overview of the lidar project Outcome 2 - how the datasets will help to achieve the TCAP goals. Outcome 3 – which users and software to achieve the TCAP goals.	